

# A-MES

AUTOMOTIVE MES · SEOYON E-HWA

V O L 0 4

## Production Planning (PP)

Forecast · MRP · Work Order · Calendar · OEE

Document · L3 Detailed Design Specification

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Reference · Plant A · SAV Detroit, MI | Plant B · GEO Birmingham, AL

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# TABLE OF CONTENTS

## Production Planning (PP)

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**PP\_Production\_Plan** A-MES · VOL.4 · PP · Production Planning

---

**PP01\_Forecast**    A-MES · VOL.4 · PP-01 · Forecast Management · 상세 설계서

---

**PP02\_SAP\_Import** A-MES · VOL.4 · PP-02 · SAP B1 Supply Plan Import · 상세 설계서

---

**PP03\_Plan\_Confirm** A-MES · VOL.4 · PP-03 · Plan & Confirm · 상세 설계서

---

**PP04\_Work\_Order** A-MES · VOL.4 · PP-04 · Work Order Management · 상세 설계서

---

**PP05\_MRP**            A-MES · VOL.4 · PP-05 · MRP — Material Requirements Planning · 상세 설계서

---

**PP06\_Purchase\_Req** A-MES · VOL.4 · PP-06 · Purchase Requisition → SAP B1 · 상세 설계서

---

**PP07\_WO\_Release** A-MES · VOL.4 · PP-07 · Work Order Release · 상세 설계서

---

**PP\_CAL\_Calendar** A-MES · VOL.4 · PP-CAL · Production Calendar · 상세 설계서

---

**PP\_OTD\_Delivery** A-MES · VOL.4 · PP-OTD · On-Time Delivery KPI · 상세 설계서

---

**PP\_LSB\_Line\_Schedule\_Board** A-MES · VOL.4 · PP-LSB · Line Schedule Board · 상세 설계서

---

**PP\_ODM\_Operating\_Downtime\_Monitor** A-MES · VOL.4 · PP-ODM · Operating / Downtime Monitor · 상세 설계서

---

**PP\_OEE\_Line\_Availability** A-MES · VOL.4 · PP-OEE · Line Availability & OEE · 상세 설계서

---

**PP\_DTL\_Downtime\_Log** A-MES · VOL.4 · PP-DTL · Downtime Log Entry · 상세 설계서

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# PRODUCTION PLANNING

생산 계획 — Forecast · SAP B1 Sync · MRP · Auto Purchase Req. · WO Release

Full production planning cycle from customer demand intake to shop-floor release. **Pipeline:** customer Forecast (SCM portal) → SAP B1 supply plan import → MES production plan → MRP material requirements → Purchase Requisition (SAP B1 Service Layer API) → Work Order release → POP shop-floor (INJ/IMG/PNT). Dual MRP: A-MES for real-time WO blocking, SAP B1 for mid/long-term procurement.

고객 수요 수신부터 현장 발행까지 생산계획 전 사이클. **파이프라인:** 고객 Forecast (SCM 포탈) → SAP B1 공급계획 Import → MES 생산계획 수립 → MRP 자재 소요 → 자재발주 요청서 (SAP B1 Service Layer API) → WO 발행 → POP 현장 (INJ/IMG/PNT). 이중 MRP: A-MES는 실시간 WO 차단, SAP B1은 중장기 발주 담당.

| SCREENS | ERP    | CUSTOMERS | MRP           | PR INTERFACE      | ROUTING |
|---------|--------|-----------|---------------|-------------------|---------|
| 13      | SAP B1 | SAV · GEO | Dual (MES+B1) | Service Layer API | A · B   |

Click any screen card / icon / pill for L3 detailed spec · 각 화면 카드 클릭 시 L3 상세 설계서

PP-01 Forecast

PP-02 SAP B1 Import

PP-03 Plan & Confirm

PP-04 Work Order

PP-05 MRP

PP-06 Purchase Req.

PP-07 WO Release

PP-CAL Calendar

PP-0TD Delivery KPI

PP-LSB Line Schedule Board

PP-0DM Downtime Monitor

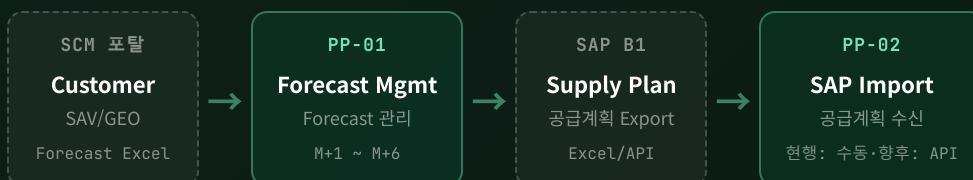
PP-0EE Line OEE

PP-DTL Downtime Log

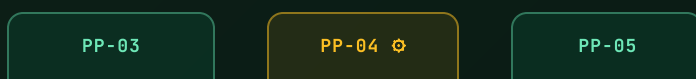
## END-TO-END FLOW · 전체 업무 흐름

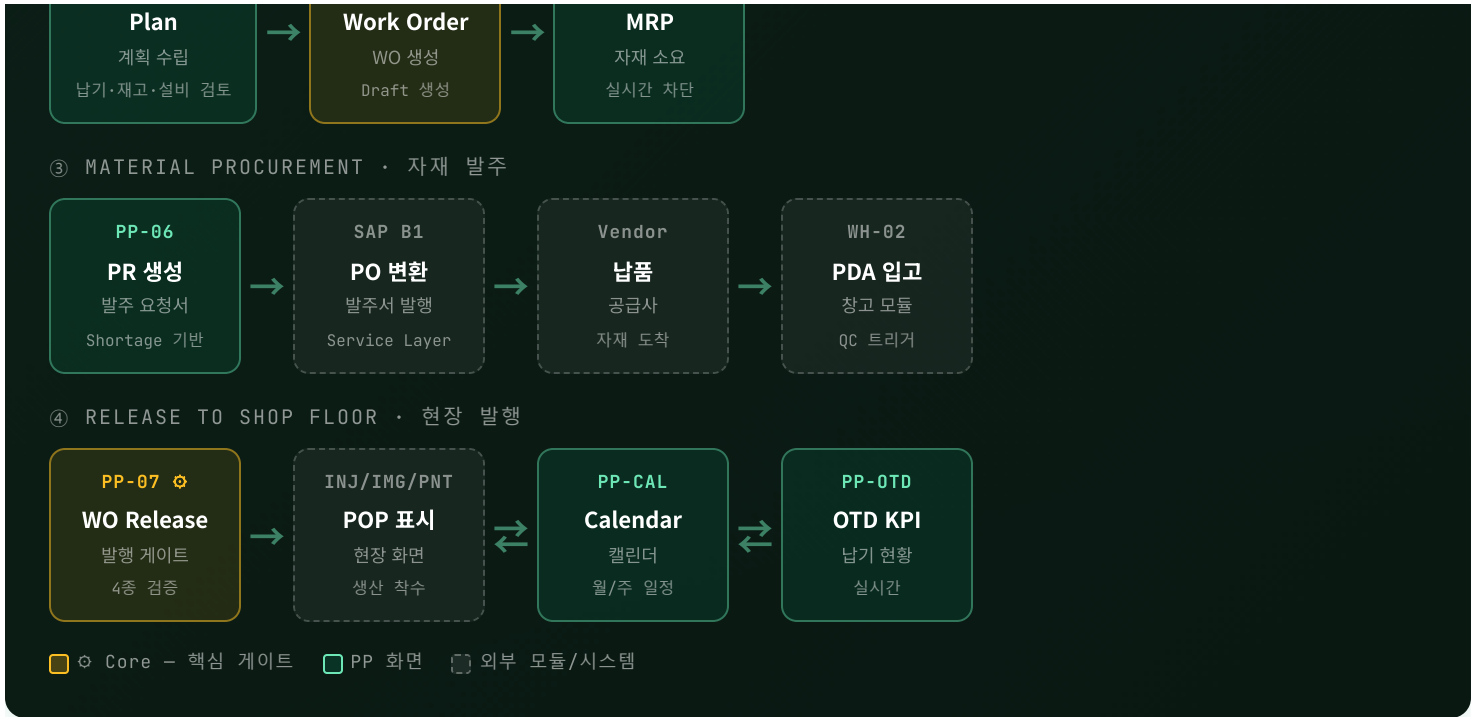
### DEMAND → PLAN → MATERIAL → RELEASE

#### ① DEMAND INTAKE · 수요 수신



#### ② PLAN & WO CREATION · 계획·WO 생성

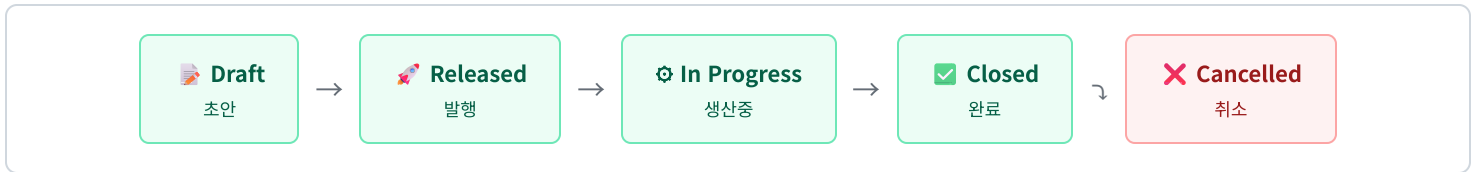




▼ MODULE KPI · 핵심 지표



▼ WO STATUS FLOW · WO 상태 흐름



▼ 9 SCREENS · 화면 설계

PP-01  
Customer Forecast Management  
고객사 Forecast 수신 및 관리

Web PC  
1st Dev

L3 DETAIL →



What you'll see: 고객 SCM 포탈 Forecast Excel 다운로드 + A-MES Import + 품번 매핑 화면  
Forecast 대비 확정 수주 편차 자동 추적 (Gap > 20% 시 PLAN/MGR 알림)





INPUT

SCM Excel (SAV/GEO)



OUTPUT

Forecast Table + Gap KPI



HORIZON

M+1 ~ M+6



ALERT

Gap &gt; 20%

## EN · DESCRIPTION

Receives and manages customer demand forecasts (M+1 to M+6) from SAV/GEO SCM Portal. Downloads Forecast Excel and imports into A-MES with item mapping. Tracks gap between forecast and confirmed orders. Forecast re-upload retains history. Unmapped customer item number triggers warning.

## KO · 설명

고객사(SAV/GEO) SCM 포탈에서 Forecast Excel을 다운로드하여 A-MES에 Import. 품번 매핑 후 Forecast 대비 확정 수주 편차 추적. Forecast 재업로드 시 이력 보관 후 UPDATE. 고객사 품번 미매핑 시 경고.



## Manual Import

고객사 포탈에서 Excel 다운로드 후 업로드.  
매월 정기



## Item Mapping

고객 품번 ↔ MD-01 자사 품번 매핑.  
미매핑 경고 (BR-F001)



## Gap Tracking

Forecast vs 확정 수주 편차 자동 계산.  
>20% 알림 (BR-F002)



## History Retention

재업로드 시 이력 보관 후 UPDATE.  
PP\_ForecastHistory (BR-F003)

관련 BR ▶

BR-PP-F001

BR-PP-F002

BR-PP-F003

PP-02

## SAP B1 Supply Plan Import

SAP B1 공급계획 수신

Web PC

SAP B1

L3 DETAIL →



What you'll see: SAP B1 확정 공급계획(판매오더) Import 화면 — 현행 Excel 수동 / 향후 Service Layer API 자동  
PP-03 생산계획 수립의 입력 데이터



INPUT

SAP B1 Excel / API



OUTPUT

Supply Plan Table → PP-03



CYCLE

현행 일 1회 · 향후 1h



INTEGRATION

Service Layer (planned)

## EN · DESCRIPTION

Receives confirmed supply plans (sales orders) from SAP Business One. **Current:** SAP B1 Excel Export → A-MES manual Import. **Future roadmap:** SAP B1 Service Layer REST API auto-polling every 1 hour. Drives PP-03 production planning.

## KO · 설명

SAP Business One의 확정 공급계획(판매오더)을 A-MES로 수신. **현행:** SAP B1 Excel Export → A-MES 수동 Import. **향후:** SAP B1 Service Layer REST API 자동 폴링(1시간)으로 전환 예정. PP-03 생산계획 수립 입력.



Current: Excel



Future: API Poll



Diff Detect

SAP B1 Export Excel → 수동 Import.  
현 운영 방식

Service Layer REST API 자동 폴링.  
1시간 간격 예정

이전 Import 대비 변경 라인 강조.  
PLAN 검토 효율



### → PP-03

확정 시 PP-03 생산계획으로 인계.  
자동 분기

관련 BR ▶ 참조: PP-03 plan rules



PP-03

## Production Planning & Confirmation

생산계획 수립 및 확인

Web PC

1st Dev

L3 DETAIL →



**What you'll see: PP-02 수신 공급계획 + FG 재고 + 설비 부하 + Forecast 비교 화면 — 4 변수 한 번에 검토**  
PLAN 담당자가 생산 가능 여부 확정 → WO 일괄 생성 대상 마킹

INPUT

PP-02 Plan + FG + Equip

OUTPUT

Confirmed Plan → PP-04

ROLE

PLAN Manager

UI

4-Variable Compare

EN · DESCRIPTION

PLAN manager reviews received supply plans against current FG inventory, equipment load, and Forecast reference. Confirms feasible items for batch WO creation in PP-04. Out-of-capacity items are flagged with overflow indicator for re-scheduling.

KO · 설명

PLAN 담당자가 PP-02 공급계획을 완제품 재고·설비 부하·Forecast 참고치와 대조하여 생산 가능 여부를 검토하고 PP-04 WO 일괄 생성 대상을 확정. 캐파 초과 품목은 overflow 표시로 재배정 권유.



### FG Stock Check

현 FG 재고 vs 출하 요구.  
잉여 자동 차감



### Capacity Load

설비/라인 가용 캐파 vs 계획 부하.  
초과 빨강 표시



### Forecast Reference

PP-01 Forecast 참고 표시.  
트렌드 검증



### → PP-04 Batch

확정 행 일괄 WO 생성 트리거.  
Ready 토글

관련 BR ▶ 참조: PP-04 (WO 생성), PP-05 (MRP)



PP-04 · CORE

## Work Order Management

Web PC

Core

L3 DETAIL →



**What you'll see: WO 4-상태 (Draft → Released → In Progress → Closed) 그리드 + 라우팅 A/B 배지 + D-Day + 진**

**행률 + SAP B1 링크**

Phase-0 미완료 품목은 자동 차단 (BR-PP-001)

INPUT

PP-03 Confirmed

OUTPUT

WO Draft → MRP/Release

STATES

Draft-Released-InProg-Closed

BLOCK

Phase-0 미완료

EN · DESCRIPTION

Manages Work Orders from Draft through Released, In Progress, to Closed. Displays routing type badge (A/B), progress bar, D-Day indicator, and SAP B1 supply plan link. Phase-0 incomplete items are blocked (BR-PP-001). Reserved qty auto-released on WO cancel.

KO · 설명

WO를 Draft → Released → In Progress → Closed 상태로 관리. 라우팅 A/B 배지·진행률·납기 D-Day·SAP B1 공급계획 연결 표시.

Phase-0 미완료 품목은 자동 차단(BR-PP-001). 취소 시 예약 수량 자동 해제.



**Routing A/B Badge**

감싸기(A) / 도색(B) 시각 구분.  
MD-04 참조



**Progress Bar**

실시간 진행률 (POP 입력 연동).  
INJ/IMG/PNT



**D-Day Indicator**

납기 잔여일 + D-3 강조(BR-008).  
캘린더 빨강



**Phase-0 Gate**

마스터 미완료 차단 (BR-001).  
Item/BOM/BOP 필수

관련 BR ▶

BR-PP-001

BR-PP-004

BR-PP-008



PP-05

## MRP — Material Requirements Planning

자재 소요 계획 — 이중 MRP (MES + SAP B1)

Web PC

Batch

L3 DETAIL →



**What you'll see: BOM 기반 자재 소요량 계산 결과 + WO별 부족 자재 리스트 + 자동 차단 표시 + 야간 배치 이력**

A-MES MRP: 실시간 WO 차단 / SAP B1 MRP: 중장기 발주 (역할 분리)

INPUT

WO + BOM + WH Stock

OUTPUT

Shortage Qty → PP-06

SCHEDULE

매일 00:00 + On-Demand

BLOCK

Shortage > 0 → WO 차단

EN · DESCRIPTION

Runs MRP to calculate material shortage quantities. **A-MES MRP**: real-time WO blocking, immediate availability check.

KO · 설명

MRP를 실행하여 자재 부족 수량 계산. **A-MES MRP**: 실시간 WO 차단, 즉시 가용성 확인. **SAP B1 MRP**: 중장기 자재 소요·공급업체 발주 담

**SAP B1 MRP:** mid/long-term procurement planning, PO generation. Dual MRP runs in parallel with clear role separation. Nightly batch at 00:00 recomputes all shortages.

당. 두 MRP 역할 분리하여 병행 운영. 매일 00:00 배치 전체 재계산.



#### A-MES Realtime

WO 생성 즉시 ShortageQty 계산.  
BR-PP-002 차단



#### B1 Mid-Term

SAP B1이 중장기 발주 계획.  
PO 생성



#### Nightly Batch

매일 00:00 전체 재계산.  
BR-PP-005



#### → PP-06 PR

Shortage → PR 자동 생성 대상.  
자동 분기

관련 BR ▶

BR-PP-002

BR-PP-005



PP-06

## Purchase Requisition → SAP B1

자재발주 요청서 (PR) → SAP B1 전송

Web PC

SAP B1

L3 DETAIL →



**What you'll see:** MRP 부족 자재 기반 PR 생성 폼 + SAP B1 전송 버튼 + 전송 결과 DocNum 수신 + 실패 시 재시도  
Service Layer API (POST /PurchaseRequests) · 실패 시 3회 재시도

INPUT

MRP Shortage

TARGET

SAP B1 PR (DocNum)

API

Service Layer REST

RETRY

3회 자동

EN · DESCRIPTION

Generates Purchase Requisition records from MRP shortage results and transmits them to SAP B1 via Service Layer API (POST /PurchaseRequests). Tracks SAP B1 DocNum after successful transmission. 3 auto-retries on failure; persistent failure escalates to PLAN team.

KO · 설명

MRP 부족 자재를 기반으로 발주요청서(PR)를 생성하고, SAP B1 Service Layer API(POST /PurchaseRequests)로 전송. 전송 성공 시 SAP B1 DocNum 수신·저장. 실패 시 3회 자동 재시도; 지속 실패 시 PLAN 팀 알림.



#### Auto Generate

MRP Shortage → PR row 자동 생성.  
수량·납기 자동



#### SAP B1 API

Service Layer POST 전송.  
DocNum 수신



#### Retry ×3

전송 실패 시 자동 재시도.  
BR-PP-006



#### Failure Alert

3회 모두 실패 시 PLAN 팀 알림.



PP-07 · ⚙ CORE

## Work Order Release

WO 발행 및 생산 착수 — 4종 검증 게이트

Web PC

Phase 0 + 1st

L3 DETAIL →

What you'll see: WO Released 전환 전 4종 조건 검증 화면 (Phase-0·MRP·설비·납기) + 통과 시 자동 자재 예약 + POP



### 노출

발행 핵심 게이트 — 한 가지라도 실패하면 차단

INPUT

WO Draft

OUTPUT

WO Released → POP

CHECKS

4 (P0·MRP·Equip·납기)

⚡ AUTO

자재 예약 + 자동 노출

EN · DESCRIPTION

Validates all release conditions before transitioning WO to Released status: **(1) Phase-0 complete, (2) MRP no shortage, (3) equipment available, (4) delivery lead time sufficient.** All pass → WO release → auto material reservation → POP screen exposure (INJ/IMG/PNT).

KO · 설명

WO Released 전환 전 4가지 조건 순서대로 검증: **(1) Phase-0 완료, (2) MRP 자재 충족, (3) 설비 가용, (4) 납기 여유.** 모두 통과 시 WO 발행 → 자재 자동 예약 → POP 화면 노출(INJ/IMG/PNT).



### 4-Check Gate

순차 검증, 1개라도 실패 시 차단.  
P0·MRP·설비·납기



### Auto Reserve

발행 시 자재 자동 예약.  
BR-PP-003



### Equip Warning

설비 Blocked 시 경고 + 보류 권고.  
BR-PP-007



### POP Exposure

발행 즉시 현장 POP 표시.  
INJ/IMG/PNT



PP-CAL

## Production Calendar

생산 캘린더 — 월/주 시각화

Web PC

2nd Dev

L3 DETAIL →



**What you'll see: 전체 WO를 월/주 캘린더로 시각화 — 색상 = 라우팅·고객사, 빨강 = D-3 임박**  
드래그로 일정 재배정 (라인장 PIN 필요)

INPUT

All WOs (PP-04)

OUTPUT

Visual Schedule + Drag

UI

Month / Week Toggle

HIGHLIGHT

D-3 빨강

EN · DESCRIPTION

Visual monthly/weekly calendar of all Work Orders with due dates. D-3 items are highlighted in red. Drag-to-reschedule supported (supervisor PIN required). Filters by routing (A/B), customer (SAV/GEO), and status. Click event to drill into PP-04 WO detail.

KO · 설명

전체 WO를 월/주 단위 캘린더로 시각화. D-3 이내 항목 빨간색 강조. 드래그 일정 조정 지원(라인장 PIN). 라우팅·고객사·상태 필터. 이벤트 클릭 시 PP-04 WO 상세 진입.



**Month / Week View**

월/주 단위 토클.  
FullCalendar.js



**Color Code**

라우팅 A/B + 고객사 SAV/GEO 색상.  
시각 분류



**D-3 Red**

납기 임박 빨간 강조 (BR-008).  
농침 방지



**Drag Reschedule**

드래그로 일정 변경 (PIN).  
감사 기록

관련 BR ▶

BR-PP-008



PP-0TD

**On-Time Delivery KPI**

납기 준수율 현황 — 실시간 대시보드

Web PC

2nd Dev

L3 DETAIL →



**What you'll see: OTD 대시보드 — 고객사·라우팅·기간별 정시/지연 비율 + 지연 원인 분석 (자재·설비·품질)**  
실시간 갱신, 95% 목표 대비 실적 추적

INPUT

FG Shipment + WO

OUTPUT

KPI Dashboard + Cause

TARGET

≥ 95%

DRILL

지연 원인 자동 분류

EN · DESCRIPTION

Real-time OTD KPI dashboard by customer (SAV/GEO), routing (A/B), and time period (day/week/month). Shows on-time vs delayed deliveries with root-cause attribution (material

KO · 설명

고객사(SAV/GEO)·라우팅(A/B)·기간(일/주/월)별 OTD 실시간 대시보드. 정시/지연 비율, 지연 원인(자재 부족·설비 정지·품질 재작업 등) 분석 제공. 목표 ≥ 95%. 지연 WO 목록 드릴다운.

shortage, equipment downtime, quality rework, etc.). Target  $\geq$  95%. Drill-down to delayed WO list.



#### Multi-Dimension

고객·라우팅·기간 3축 필터.  
조합 분석



#### Root Cause

지연 원인 자동 분류·집계.  
자재/설비/품질



#### Trend Chart

시계열 OTD 추이.  
목표선 95%



#### Delay Drill

지연 WO 목록 드릴다운.  
PP-04 진입

관련 BR ▶ 참조: BR-PP-008 (D-3 알림)



PP-LSB

## Line Schedule Board

라인별 분단위 스케줄 보드

Web PC

2nd Dev

L3 DETAIL →



**What you'll see:** WO를 각 라인의 고유 시간패턴(MD-29) 위에 배치하는 분단위 간트 — 가동/계획비가동 밴드 + 캐파 부하  
올  
가동창 초과·계획비가동 침범 시 배치 차단

INPUT

MD-29 Pattern + WO

OUTPUT

PP\_LineSchedule + POP

GRAIN

1-minute slot

AUTO

Auto-fill by D-Day

EN · DESCRIPTION

Minute-level Gantt board scheduling Work Orders onto each line's own time pattern. Lines render their MD-29 operating / planned-down bands; WO blocks drop only onto operating windows. Capacity =  $\Sigma$  operating minutes; overflow blocks Publish.

KO · 설명

작업지시를 라인별 고유 시간패턴 위에 배치하는 분단위 간트 보드. MD-29 가동·계획비가동 밴드 렌더링, WO 블록은 가동창에만 배치. 캐파 =  $\Sigma$ 가동분, 초과 시 발행 차단.



#### Per-line Pattern

라인별 MD-29 시간패턴 기준.  
공용 캘린더 아님



#### Window Fit

WO는 단일 가동창 내 배치.  
BR-PP-031



#### Capacity Load

$\Sigma$ 가동분 대비 부하율.  
100% 초과 차단



#### Publish

POP 푸시 + ODM 기준선.  
BR-PP-033

관련 BR ▶

BR-PP-030

BR-PP-031

BR-PP-032



PP-ODM

## Operating / Downtime Monitor

가동·계획비가동·실제비가동 실시간 모니터

Web PC / 현황판

2nd Dev

L3 DETAIL →



**What you'll see:** 라인별 계획 패턴(MD-29) vs 실제 설비 신호를 실시간 대조 — 매 분(分)을 가동/계획비가동/실제비가동 3 상태로 분류

계획상 가동인데 정지 시 UNPLANNED 자동 판정 + 사유 입력 요청

INPUT

MD-29 + 설비 신호

OUTPUT

PP\_LineStateLog

REFRESH

WebSocket 1-min

ALERT

비가동 10분 초과

EN · DESCRIPTION

Real-time wallboard comparing each line's planned MD-29 pattern against actual equipment signal. Each minute resolves to OPERATING / PLANNED\_DOWN / UNPLANNED\_DOWN; a stop inside an operating window auto-classifies as unplanned and prompts a PP-DTL reason.

KO · 설명

라인별 계획 MD-29 패턴과 실제 설비 신호를 실시간 대조하는 현황판. 매 분을 가동/계획비가동/실제비가동으로 판정, 가동창 내 정지는 실제 비가동 자동분류 + PP-DTL 사유 입력 요청.



### 3-State Classify

가동·계획비가동·실제비가동.  
분단위 판정



### Unplanned Detect

가동창 내 정지 자동분류.  
BR-PP-035



### Reason Prompt

미입력 비가동 PP-DTL 연계.  
사유 강제



### Operating Rate

가동률 실시간 집계.  
PP-OEE 피드

관련 BR ▶

BR-PP-035

BR-PP-036

BR-PP-037



PP-OEE

## Line Availability & OEE

라인 가용성·OEE 롤업

Web PC

2nd Dev

L3 DETAIL →



**What you'll see:** PP\_LineStateLog를 시간손실 워터폴로 롤업 — 부하시간 → 가동 → OEE = 가용성×성능×품질  
계획/실제 비가동을 분리해 라인 고유 패턴을 공정 반영





INPUT

PP\_LineStateLog



OUTPUT

PP\_LineOEE snapshot



PERIOD

Day / Week / Month



TARGET

OEE ≥ 75%

## EN · DESCRIPTION

Rolls up the classified state log into a time-loss model and computes per-line OEE. Loading Time minus planned + unplanned downtime gives Availability; combined with Performance and Quality it yields  $OEE = A \times P \times Q$ , snapshotted per period.

## KO · 설명

분류된 상태 로그를 시간손실 모델로 롤업하여 라인별 OEE 산출. 부하시간 - 계획·실제 비가동 → 가용성, 성능·품질과 결합해  $OEE = A \times P \times Q$ , 기간별 스냅샷 저장.



## Loss Waterfall

부하→가동→가치가동 손실 분해.  
시각 분석



## Planned Excluded

가용성은 실제 비가동만 손실.  
BR-PP-041



## Period Snapshot

기간별 OEE 고정 스냅샷.  
감사 추적



## RPT Feed

RPT-06/07로 전달.  
OEE·KPI

관련 BR ▶

BR-PP-042

BR-PP-041

BR-PP-044

## PP-DTL



## Downtime Log Entry

비가동 로그 입력 — POP 터치패널

POP / PDA

2nd Dev

L3 DETAIL →



What you'll see: PP-ODM이 감지한 실제 비가동 이벤트에 작업자가 사유(MD-25)·4M1E 원인(MD-22)을 터치 입력

사유 기록 후 이벤트 종료 → PP-OEE 가용성 손실 귀속



INPUT

ODM Downtime Event



OUTPUT

PP\_LineDowntimeLog



REASON

MD-25 · MD-22



DEVICE

POP 15" 터치

## EN · DESCRIPTION

POP touch-panel screen where the operator records the reason for an unplanned downtime event detected by PP-ODM. Taps a downtime reason (MD-25) and optional 4M1E cause (MD-22); the closed event flows to PP-ODM and PP-OEE.

## KO · 설명

PP-ODM이 감지한 실제 비가동 이벤트의 사유를 작업자가 기록하는 POP 터치패널. 비가동 사유(MD-25)·4M1E 원인(MD-22) 터치 선택, 종료된 이벤트는 PP-ODM·PP-OEE로 전달.



## Reason Tiles

MD-25 DOWNTIME 사유 터치.



## 4M1E Cause

MD-22 원인 분류 선택.



## Comment Gate

필수 사유는 코멘트 강제.

대형 타일

선택 입력

BR-PP-046

**MNT Sync**

고장 사유는 MNT-06 동기화.

BR-PP-048

관련 BR ▶

BR-PP-045

BR-PP-046

BR-PP-048

## ▼ BUSINESS RULES · 업무 규칙 (PP · 16건)

| Rule ID    | Rule (EN)                                         | 규칙 (KO)                        | 조건 · Condition             | Action |
|------------|---------------------------------------------------|--------------------------------|----------------------------|--------|
| BR-PP-F001 | Warn on unmapped customer item no.                | 고객사 품번 미매핑 시 경고                | CustomerItemNo 매핑 없음       | Warn   |
| BR-PP-F002 | Alert on Forecast vs Order gap > 20%              | Forecast 대비 수주 편차 20% 초과 알림    | Gap > Forecast × 0.2       | Alert  |
| BR-PP-F003 | Retain history on Forecast re-upload              | Forecast 재업로드 시 이력 보관 후 UPDATE | 동일 월·품목 중복                 | Auto   |
| BR-PP-001  | Block WO if Phase-0 incomplete                    | Phase-0 미완료 WO 생성 차단           | Phase0Complete = 0         | Block  |
| BR-PP-002  | Block WO on MRP shortage                          | MRP 재고 부족 WO 자동 차단             | ShortageQty > 0            | Block  |
| BR-PP-003  | Auto-reserve material on WO release               | WO 발행 시 자재 자동 예약               | Status = Released          | Auto   |
| BR-PP-004  | Release reserved qty on WO cancel                 | WO 취소 시 예약 수량 자동 해제            | Status = Cancelled         | Auto   |
| BR-PP-005  | MRP auto-run nightly at 00:00                     | MRP 야간 자동 실행                   | 매일 00:00 배치                | Auto   |
| BR-PP-006  | Retry PR SAP B1 transmission ×3                   | PR SAP B1 전송 실패 시 3회 재 시도      | 전송 오류                      | Retry  |
| BR-PP-007  | Warn if equipment blocked on WO release           | WO 발행 전 설비 가용 상태 확인            | Equipment.Status = Blocked | Warn   |
| BR-PP-008  | Alert DDay ≤ 3 WO                                 | 납기 D-3 이내 WO 강조 알림             | DueDate - TODAY ≤ 3        | Alert  |
| BR-PP-030  | Each line schedules on its own MD-29 time pattern | 각 라인은 고유 MD-29 시간패턴으로 스케줄링     | PP-LSB 일정 수립               | Auto   |
| BR-PP-031  | WO block must fit one OPERATING window            | WO 블록은 단일 가동창 내 완전 배치          | 가동창 초과·침범                  | Block  |
| BR-PP-035  | Stop in operating window → UNPLANNED_DOWN         | 가동창 내 설비 정지는 실제비가동 자동 분류       | 설비 신호 STOP                 | Auto   |
| BR-PP-042  | OEE = Availability × Performance × Quality        | OEE = 가용성 × 성능 × 품질, 기간별 스냅샷   | PP-OEE 롤업                  | Auto   |
| BR-PP-045  | Unplanned downtime must close with a reason code  | 실제 비가동은 사유코드로 종료 필수            | 비가동 이벤트 종료                 | Block  |



VOL.4 · PP · PRODUCTION PLANNING · FORECAST 관리

# PP-01 FORECAST MANAGEMENT

FORECAST

DEVICE  Web PC

PHASE 1st Dev

ROLE PLAN Manager

FREQ Monthly · 월간

URL /pp/forecast

## 01 Overview

화면 개요 · 목적 · 컨텍스트

Manages customer forecast intake (SAV/GEO). Excel files downloaded from customer SCM portals are uploaded, validated, and mapped against the PP\_CustomerItemMap table to convert customer item numbers to A-MES item codes. Provides M+1 to M+6 monthly demand visibility with M+1~M+3 binding for production planning and M+4~M+6 as reference for capacity planning. Detects gaps between forecast and confirmed orders, alerting PLAN/MGR when deviation exceeds 20%.

고객사(SAV/GEO) Forecast 수신·관리 화면. 고객 SCM 포털에서 다운로드한 Excel을 업로드하면 PP\_CustomerItemMap으로 고객사 품번 → A-MES 품목코드 매핑·검증. M+1~M+3은 생산계획 구속력, M+4~M+6은 능력계획 참고. Forecast 대비 확정 수주 편차 20% 초과 시 PLAN/MGR 알림.

|           |                                                       |
|-----------|-------------------------------------------------------|
| SCREEN ID | PP-01                                                 |
| LAYOUT    | Upload + Forecast table + Gap analysis 업로드·표·편차분석     |
| SOURCE    | SCM portal Excel (manual) SCM 포털 Excel                |
| FUTURE    | SCM Portal API (Phase 2)                              |
| TABLES    | PP_Forecast , PP_ForecastHistory , PP_CustomerItemMap |
| RBAC      | PP.Forecast                                           |

## 02 Screen Mockup

화면 목업 — 주요 UI 구성

PP-01 · FORECAST MANAGEMENT · WEB PC



A-MES · PP · Forecast Management

PP &gt; Forecast

All Customers2025-Q4Item Code...SearchUpload ExcelExport

142Mapped Items매핑 품목

3Unmapped미매핑

8Gap >20%편차 20% ↑

2025-08-01Last Upload최근 업로드

| Customer<br>고객사 | Customer Item No.<br>고객사 품번 | A-MES Code<br>내부 품목 | Item Name<br>품목명 | M+1   | M+2   | M+3   | M+4~6<br>참고 | Confirmed<br>확정 수주 | Gap<br>편차 |
|-----------------|-----------------------------|---------------------|------------------|-------|-------|-------|-------------|--------------------|-----------|
| SAV             | SAV-TRIM-LH                 | TRIM-LH-001         | 도어트림 LH          | 2,400 | 2,200 | 2,000 | 참고          | 2,350              | +2%       |
| GEO             | GEO-TRIM-RH                 | TRIM-RH-001         | 도어트림 RH          | 1,600 | 1,500 | 1,400 | 참고          | 1,580              | -1%       |
| SAV             | SAV-CONS-UP                 | CONS-UP-001         | 콘솔어퍼             | 800   | 900   | 1,000 | 참고          | 650                | -19%      |
| GEO             | GEO-NEW-PT                  | — △                 | UNMAPPED · 매핑 필요 | 500   | 500   | 500   | 참고          | —                  | N/A       |

03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트         | TYPE   | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증 · 동작                                                  | DEFAULT   |
|--------|---------------------------|--------|------------|-----------------------------------------------------------------------------------|-----------|
| CMP-01 | Customer Filter<br>고객사 필터 | SELECT | N          | Options: All / SAV / GEO. Affects all downstream listings.                        | All       |
| CMP-02 | Period Selector<br>기간 선택  | SELECT | N          | Quarter/Month selector. Loads forecast for selected period.                       | Current Q |
| CMP-03 | Upload Excel<br>Excel 업로드 | FILE   | Y          | Accepts .xlsx with required columns: Customer Item No, M+1, M+2, M+3, M+4~6.      | —         |
| CMP-04 | Validate Mapping<br>매핑 검증 | AUTO   | —          | Auto-runs on upload. Joins with PP_CustomerItemMap . Unmapped rows flagged.       | —         |
| CMP-05 | Gap Calculation<br>편차 계산  | AUTO   | —          | $Gap\% = (Confirmed - Forecast) / Forecast \times 100$ . >20% → BR-PP-F002 alert. | —         |
| CMP-06 | Export Excel<br>Excel 출력  | BUTTON | N          | Exports current view with gap analysis.                                           | —         |

Event & Action Flow

1

**USER** click · [📁 Upload Excel]

User selects Excel file from customer SCM portal.

고객 SCM 포탈에서 다운로드한 Excel 선택.

**VALIDATE** Schema check: required columns present, no empty Item No.

2

**SYSTEM** parse · file → rows

System parses rows and matches against PP\_CustomerItemMap for item code mapping.

행을 파싱하여 PP\_CustomerItemMap으로 품번 매핑.

**EFFECT** Mapped: insert/update `PP_Forecast`. Unmapped: flag in warning list.

3

**SYSTEM** calc · gap analysis

For each mapped item, fetch confirmed orders from PP\_SupplyPlan and compute gap%.

매핑된 품목별로 PP\_SupplyPlan의 확정 수주와 비교하여 편차% 계산.

**ALERT** Gap > 20% → BR-PP-F002 alert sent to PLAN/MGR via email.

4

**SYSTEM** history · re-upload

If duplicate month+item rows, move existing to PP\_ForecastHistory before update.

동일 월·품목 중복 시 기존 행을 PP\_ForecastHistory로 이관 후 갱신.

**AUDIT** History retained 3 years per audit policy.

5

**USER** click · row [Map] (for unmapped)

User clicks Map button on unmapped row, opens dialog to manually link customer item no to A-MES item code.

미매핑 행의 [Map] 클릭 시 수동 매핑 다이얼로그 표시.

**API** `POST /api/pp/customer-item-map` · adds mapping for future uploads.

6

**SYSTEM** complete · forecast saved

Forecast tab refreshed. Downstream PP-03 plan creation can now reference these qty.

Forecast 탭 갱신. PP-03 생산계획 수립에서 이 수량 참조 가능.

**DOWNSTREAM** PP-03 (planning), PP-05 (MRP capacity check)

## Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP01-001

Unmapped Item

미매핑 품번

|         |                                                                 |
|---------|-----------------------------------------------------------------|
| CAUSE   | Customer item no not found in <code>PP_CustomerItemMap</code> . |
| DETECT  | On upload parse.                                                |
| ACTION  | Row flagged red, excluded from save. Warning list shown.        |
| RECOVER | User maps via [Map] button or MD team updates mapping master.   |

⚠ UNMAPPED · MAP REQUIRED

E-PP01-002

### Invalid File Format

파일 형식 오류

|         |                                                      |
|---------|------------------------------------------------------|
| CAUSE   | Excel missing required columns or wrong type.        |
| DETECT  | Schema validation on upload.                         |
| ACTION  | Reject entire upload. Show specific missing columns. |
| RECOVER | Use [Sample Template] download.                      |

⊖ INVALID FORMAT

E-PP01-003

### Forecast Gap Alert

편차 알림

|         |                                                 |
|---------|-------------------------------------------------|
| CAUSE   | Confirmed orders vs forecast deviation > 20%.   |
| DETECT  | Auto on save.                                   |
| ACTION  | Email alert to PLAN+MGR. Item flagged in table. |
| RECOVER | Review with sales/customer.                     |

ⓘ GAP ALERT — REVIEW

E-PP01-004

### Concurrent Upload

동시 업로드 충돌

|         |                                                      |
|---------|------------------------------------------------------|
| CAUSE   | Two users upload same month forecast simultaneously. |
| DETECT  | Row lock on PP_Forecast.                             |
| ACTION  | Second user blocked with current state shown.        |
| RECOVER | Reload and re-apply changes if still needed.         |

⚠ CONCURRENT — RELOAD

E-PP01-005

### History Rollback Required

이력 복구 필요

|         |                                                                               |
|---------|-------------------------------------------------------------------------------|
| CAUSE   | User requests rollback to previous forecast version.                          |
| DETECT  | Manual via [History] tab.                                                     |
| ACTION  | Requires manager approval. Restored from <a href="#">PP_ForecastHistory</a> . |
| RECOVER | Manager approves; new entry inserted (not in-place update).                   |

⚠ ROLLBACK · APPROVAL

#### E-PP01-006 Empty Upload

빈 파일

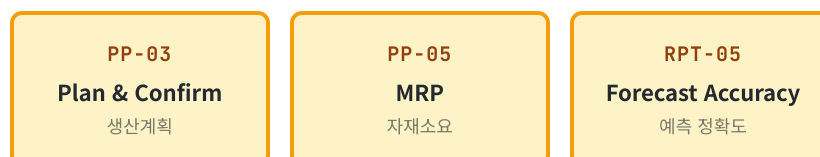
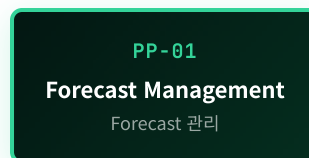
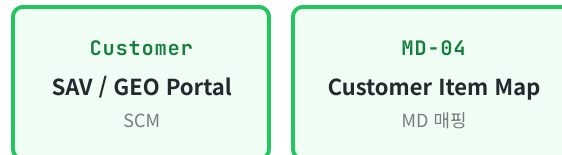
|         |                                             |
|---------|---------------------------------------------|
| CAUSE   | Excel has zero data rows.                   |
| DETECT  | Pre-parse check.                            |
| ACTION  | Block save. Suggest using correct template. |
| RECOVER | Re-download from customer portal.           |

⚠ NO DATA

## 06 Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



▼ DOWNSTREAM · 하위 화면 / 데이터 ▼

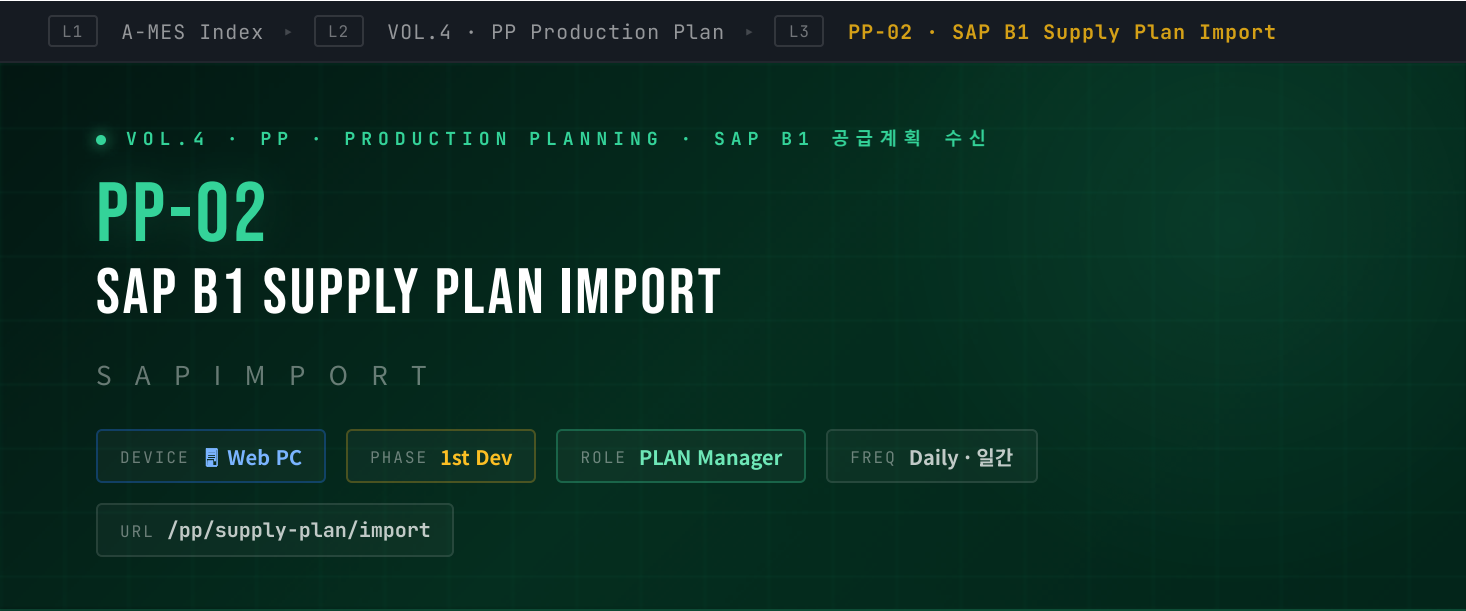


07

Applicable Business Rules

적용 업무 규칙

| RULE ID    | RULE (EN)                                                 | 규칙 (KO)                           | ACTION |
|------------|-----------------------------------------------------------|-----------------------------------|--------|
| BR-PP-F001 | Warn on unmapped customer item no. on Forecast upload.    | Forecast 업로드 시 미매핑 고객사 품번 경고.     | ALT    |
| BR-PP-F002 | Alert when Forecast vs confirmed order gap exceeds 20%.   | Forecast 대비 확정 수주 편차 20% 초과 시 알림. | ALT    |
| BR-PP-F003 | On Forecast re-upload, retain history before overwriting. | Forecast 재업로드 시 이력 보관 후 UPDATE.   | AUT    |



01

Overview

화면 개요 · 목적 · 컨텍스트

Receives confirmed supply plans from SAP Business One. Current (Phase 1): manual Excel import. Future (Phase 2): direct integration via SAP B1 Service Layer API. Each supply plan row corresponds to a confirmed sales order with order qty, due date, and customer reference. Once imported, plans flow to PP-03 for feasibility review and WO batch creation.

SAP B1에서 확정된 공급계획 수신. 1차: Excel 수동 import. 2차: SAP B1 Service Layer API 직접 연동. 각 행은 확정 판매오더(고객사·수량·납기). Import 후 PP-03에서 가용성 검토·WO 일괄 생성으로 연계.

|           |                                     |
|-----------|-------------------------------------|
| SCREEN ID | PP-02                               |
| SOURCE    | SAP B1 Excel export (Phase 1)       |
| FUTURE    | Service Layer API GET /Orders       |
| SCHEDULE  | Daily 07:00 KST batch 매일 07:00 배치   |
| TABLES    | PP_SupplyPlan , PP_SupplyPlanDetail |
| RBAC      | PP.Plan                             |

02

Screen Mockup

화면 목업 — 주요 UI 구성

● PP-02 · SAP B1 SUPPLY PLAN IMPORT · WEB PC

A-MES · PP · SAP B1 Supply Plan Import

PP > Supply Plan

Import Method:

Excel Upload (Phase 1)

파일 선택

선택된 파일 없음

Import

Sample Template

48

Today's SOs  
금일 SO

46

Confirmed  
확정

2

Open  
미확정

07:12

Last Sync  
동기화

| SO No.        | Customer<br>거래선 | Item Code   | Item Name | Order Qty | Due Date   | Status    | WO Link         |
|---------------|-----------------|-------------|-----------|-----------|------------|-----------|-----------------|
| SO-2025-08001 | SAV             | TRIM-LH-001 | 도어트림 LH   | 2,350     | 2025-09-15 | Confirmed | WO-20250801-001 |
| SO-2025-07055 | SAV             | TRIM-RH-001 | 도어트림 RH   | 1,580     | 2025-08-31 | Confirmed | WO-20250715-003 |
| SO-2025-08003 | GEO             | CONS-UP-001 | 콘솔어퍼      | 800       | 2025-09-25 | Open      | —               |

03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트           | TYPE      | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증 · 동작                                          | DEFAULT     |
|--------|-----------------------------|-----------|------------|---------------------------------------------------------------------------|-------------|
| CMP-01 | Import Method<br>수신 방식      | SELECT    | Y          | Excel Upload (Phase 1) / API (Phase 2, disabled until ready).             | Excel       |
| CMP-02 | File Picker<br>파일 선택        | FILE      | Y          | .xlsx file from SAP B1 export. Required: SO No, Customer, Item, Qty, Due. | —           |
| CMP-03 | [Import]<br>가져오기            | BUTTON    | Y          | Triggers parse + insert into PP_SupplyPlan + Detail . Audit log entry.    | —           |
| CMP-04 | Sample Template<br>샘플 양식    | BUTTON    | N          | Downloads blank Excel with required column headers.                       | —           |
| CMP-05 | Daily Auto-Sync<br>일 자동 동기화 | SCHEDULED | —          | 07:00 KST batch. Phase 2 only (API). Bypasses Excel.                      | Daily 07:00 |

04

Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

SCHEDULED

cron · 07:00 daily sync

1

[Phase 2 future] System auto-calls SAP B1 Service Layer to fetch `/Orders?filter=DocStatus eq 'O'` and inserts new SOs.

[2차 예정] 매일 07:00 SAP B1 Service Layer `/Orders` 호출하여 미확정 SO 자동 import.

API `GET /b1s/v1/Orders?$filter=DocStatus%20eq%20%27O%27`

2

USER **click** · [**↑ Import**]

Manager uploads SAP B1 Excel export. System validates schema and parses rows.

관리자가 SAP B1 Excel export 업로드. 스키마 검증 + 행 파싱.

VALIDATE Required cols: SO No, Customer, Item Code, Qty, Due Date, Status

3

SYSTEM **save** · **transactional insert**

Inserts header row in PP\_SupplyPlan and detail rows in PP\_SupplyPlanDetail atomically.

PP\_SupplyPlan 헤더 + PP\_SupplyPlanDetail 상세를 원자적으로 INSERT.

DB `BEGIN TRAN; INSERT header; INSERT details; COMMIT`

4

SYSTEM **match** · **forecast vs SO**

Cross-references confirmed SO against PP-01 Forecast for the same item/period.

확정 SO와 PP-01 Forecast를 품목·기간별로 매칭.

ALERT Gap > 20% triggers BR-PP-F002 alert (same as PP-01).

5

SYSTEM **complete** · **ready for PP-03**

Imported SOs appear in PP-03 with Forecast reference and FG stock context.

Import된 SO가 PP-03에서 Forecast 참조·FG 재고와 함께 표시.

DOWNSTREAM PP-03 plan creation, calendar overlay

## 05 Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP02-001

### Schema Mismatch

스키마 불일치

|         |                                                   |
|---------|---------------------------------------------------|
| CAUSE   | Excel column names don't match expected.          |
| DETECT  | Pre-parse header check.                           |
| ACTION  | Block import. Show missing/extra columns.         |
| RECOVER | Re-download SAP B1 export or use Sample Template. |

— COLUMN MISMATCH

E-PP02-002

### Duplicate SO No

SO 중복

|         |                                                 |
|---------|-------------------------------------------------|
| CAUSE   | SO No already exists in PP_SupplyPlan.          |
| DETECT  | Unique key constraint.                          |
| ACTION  | Show duplicate count, ask user: update or skip. |
| RECOVER | Update overwrites Qty/Due Date.                 |

⚠️ DUPLICATE SO

E-PP02-003

### SAP API Timeout

SAP API 타임아웃

|         |                                                 |
|---------|-------------------------------------------------|
| CAUSE   | [Phase 2] Service Layer call exceeds 30s.       |
| DETECT  | Heartbeat monitor.                              |
| ACTION  | Retry 3× with exponential backoff. Then alert.  |
| RECOVER | Manual Excel fallback or wait for SAP recovery. |

🚫 SAP TIMEOUT

E-PP02-004

### Unknown Item Code

미등록 품목

|         |                                                     |
|---------|-----------------------------------------------------|
| CAUSE   | SO item not in <code>MD_Item</code> master.         |
| DETECT  | Per-row validation.                                 |
| ACTION  | Skip row, log to error report. Other rows continue. |
| RECOVER | MD team adds item or removes invalid SO in SAP.     |

🚫 ITEM NOT IN MD

E-PP02-005

### Past Due Date

과거 납기

|         |                                                                                   |
|---------|-----------------------------------------------------------------------------------|
| CAUSE   | SO due date is in the past.                                                       |
| DETECT  | Date validation.                                                                  |
| ACTION  | Allow import but flag yellow. Cannot create WO from past-due SO without override. |
| RECOVER | Update due date in SAP B1.                                                        |

⚠️ PAST DUE

E-PP02-006

## Authentication Failed

인증 실패

CAUSE [Phase 2] SAP B1 Service Layer login token expired.

DETECT API 401 response.

ACTION Re-authenticate via stored credentials. Retry once.

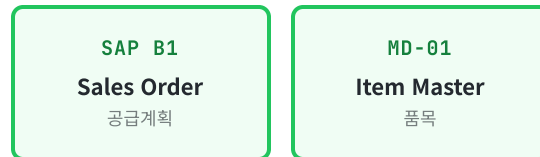
RECOVER SYS admin updates SAP credential secret.

🚫 AUTH FAILED

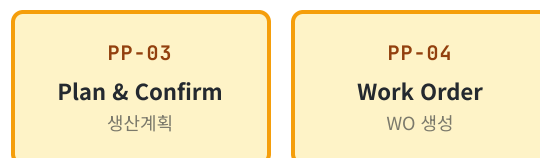
## 06 Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



PP-02  
SAP B1 Supply Plan Import  
SAP B1 공급계획 수신



▼ DOWNSTREAM · 하위 화면 / 데이터 ▼

## 07 Applicable Business Rules

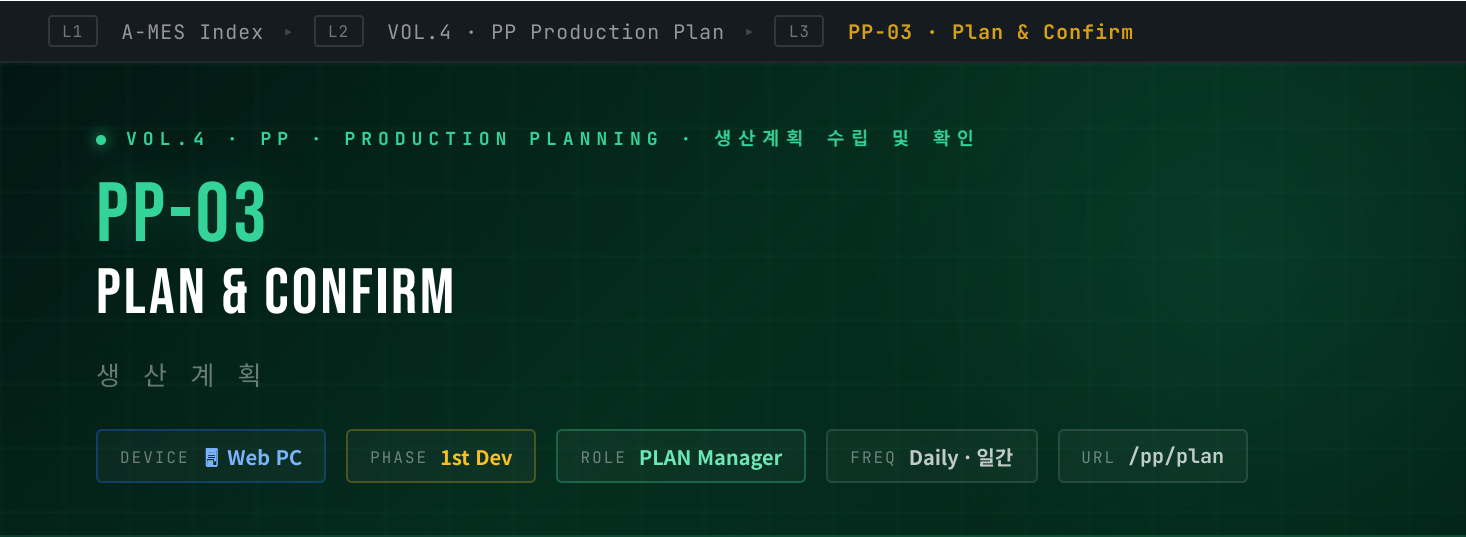
적용 업무 규칙

| RULE ID    | RULE (EN)                                               | 규칙 (KO)                           | ACTION |
|------------|---------------------------------------------------------|-----------------------------------|--------|
| BR-PP-F002 | Alert when Forecast vs confirmed order gap exceeds 20%. | Forecast 대비 확정 수주 편차 20% 초과 시 알림. | ALT    |

|           |                                                         |                              |     |
|-----------|---------------------------------------------------------|------------------------------|-----|
| BR-PP-001 | Block WO creation if Phase-0 master data is incomplete. | Phase-0 기준정보 미완료 시 WO 생성 차단. | BLK |
|-----------|---------------------------------------------------------|------------------------------|-----|

A-MES · VOL.4 · PP-02 SAP B1 Supply Plan Import · L3 Detail  
Spec v1.0 · Bilingual EN/KO

Automotive Parts MES · SAV & GEO  
Supply · Confidential



01

Overview

화면 개요 · 목적 · 컨텍스트

PLAN manager reviews supply plans against FG inventory, equipment load, and forecast reference. For each plan line, the system calculates Net Required Qty (= Order Qty – FG Stock), shows current line load %, and compares vs forecast. Items selected via checkbox are batch-released to PP-04 as Work Orders.

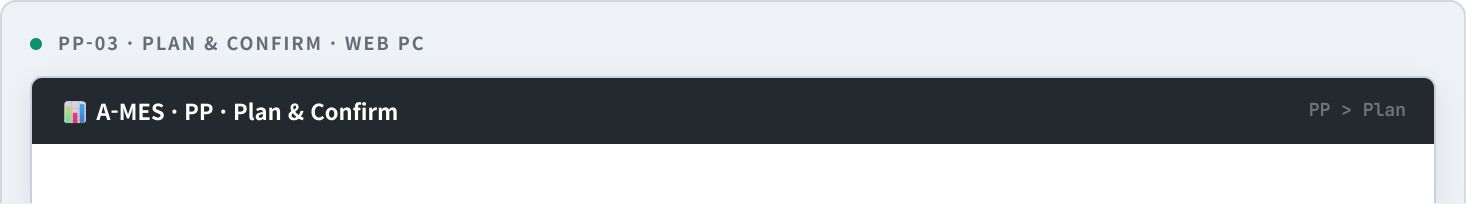
PLAN 담당자가 공급계획을 완제품 재고·설비 부하·Forecast와 대조하여 생산 가능성 검토. 행별로 순생산 필요량(Order – FG Stock) 자동 계산, 라인 부하율 표시, Forecast 대비. 체크박스로 선택된 행은 PP-04 WO 일괄 생성으로 전송.

|           |                                                  |
|-----------|--------------------------------------------------|
| SCREEN ID | PP-03                                            |
| SOURCE    | PP-02 imported supply plans                      |
| CALC      | Net Req = Order – FG Stock                       |
| CAPACITY  | MNT equipment load (real-time)                   |
| TABLES    | PP_SupplyPlan , WH_Inventory (FG), MNT_Equipment |
| RBAC      | PP.Plan                                          |

02

Screen Mockup

화면 목업 — 주요 UI 구성





2025-09-15

2025-09-30

All Lines

Search

Calendar View

Batch Create WO

|                                     | SO No.        | Item                        | Order<br>공급 | FG Stock<br>완제품 | Net Req.<br>순생산      | Forecast | Line Load<br>설비 부하         | Due Date   |
|-------------------------------------|---------------|-----------------------------|-------------|-----------------|----------------------|----------|----------------------------|------------|
| <input checked="" type="checkbox"/> | SO-2025-08001 | TRIM-LH-001<br>도어트림 LH      | 2,350       | 320             | 2,030                | 2,400    | <div><div></div></div> 72% | 2025-09-15 |
| <input type="checkbox"/>            | SO-2025-08003 | CONS-UP-001<br>콘솔어퍼         | 800         | 150             | 650                  | 900      | <div><div></div></div> 45% | 2025-09-25 |
| <input type="checkbox"/>            | SO-2025-08009 | TRIM-RH-002<br>도어트림 RH (신규) | 1,200       | 0               | 1,200 <span>⚠</span> | —        | Phase-0 미완료                | 2025-09-20 |

03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트               | TYPE     | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증 · 동작                                             | DEFAULT   |
|--------|---------------------------------|----------|------------|------------------------------------------------------------------------------|-----------|
| CMP-01 | Date Range<br>기간 선택             | DATE×2   | Y          | From-To. Default: this week + next week. Max 90 days.                        | Week+1    |
| CMP-02 | Line Filter<br>라인 필터            | SELECT   | N          | Filter by production line: INJ-L1/L2, IMG-L1, PNT-L1.                        | All       |
| CMP-03 | Plan Checkbox<br>계획 체크박스        | CHECKBOX | Y          | Per-row selection for batch WO creation. Disabled if blocking conditions.    | Unchecked |
| CMP-04 | Net Req Calc<br>순생산 계산          | AUTO     | —          | <div>Net Req = Order Qty - FG Stock</div> . Real-time from WH_Inventory.     | —         |
| CMP-05 | Line Load Display<br>설비 부하 표시   | AUTO     | —          | Bar showing current line utilization %. Red >90%, yellow 70-90%, green <70%. | —         |
| CMP-06 | [▶ Batch Create WO]<br>WO 일괄 생성 | BUTTON   | Y          | Creates Draft WOs for all checked rows. Validates Phase-0 first.             | —         |
| CMP-07 | Calendar View<br>캘린더 보기         | BUTTON   | N          | Switches to PP-CAL calendar view, preserving filter.                         | —         |

04

Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

**USER** page.load · open PP-03

Loads all confirmed SOs from PP\_SupplyPlan with status Open in date range. Calculates Net Req for each.  
기간 내 확정 SO 로드 + 행별 Net Req 자동 계산.

**API** GET /api/pp/plan?from=...&to=...

2

**SYSTEM** fetch · FG inventory

Real-time fetch from WH\_Inventory for each item's FG (finished goods) on-hand qty.  
각 품목별 완제품 재고를 WH\_Inventory에서 실시간 조회.

**QUERY** SELECT item\_code, SUM(on\_hand) FROM wh\_inventory WHERE warehouse='WH-FG' GROUP BY item\_code

3

**SYSTEM** fetch · line load

Queries MNT\_Equipment for current capacity utilization per line.  
MNT\_Equipment에서 라인별 현재 가동률 조회.

**EFFECT** Each row shows line load bar

4

**USER** check · row selection

User selects rows by checkbox. System dynamically updates batch size and total qty in footer.  
행 체크 시 일괄 처리 건수 · 총 수량 푸터에 실시간 갱신.

**UI** Footer counter updates · disable conflicting rows

5

**USER** click · [Batch Create WO]

Triggers WO creation for all checked rows. Each becomes a Draft WO with routing per BOP (MD-04).  
체크된 행에 대해 WO 일괄 생성. 각각 Draft 상태, BOP(MD-04) 기준 라우팅 자동 적용.

**API** POST /api/pp/wo/batch · returns array of wo\_ids

6

**SYSTEM** validate · Phase-0

Before WO creation, validates that all required master data is complete (BOM/BOP/Inspection Standard).  
WO 생성 전 기준정보 완성도 검증(BOM/BOP/검사 기준).

**BLOCK** If any missing → BR-PP-001 block · skip row · report

7

**SYSTEM** redirect · to PP-04

Successful batch redirects to PP-04 with new WOs highlighted.  
성공 시 PP-04로 이동하며 신규 WO 하이라이트.

**NAV** /pp/wo?highlight=WO-20250801-001,002,003

## 05

## Exception Handling

예외 처리 · 오류 코드 · 복구 절차

**E-PP03-001****Phase-0 Incomplete**

Phase-0 미완료

|         |                                                            |
|---------|------------------------------------------------------------|
| CAUSE   | Selected item has missing master data (BOM/BOP/Insp).      |
| DETECT  | Validation on Batch Create.                                |
| ACTION  | Block creation for that row. Show specific missing master. |
| RECOVER | MD team completes master in MD-01~06.                      |

 PHASE-0 INCOMPLETE**E-PP03-002****Insufficient Capacity**

능력 부족

|         |                                                              |
|---------|--------------------------------------------------------------|
| CAUSE   | Line load already > 95% for the period.                      |
| DETECT  | Capacity check on select.                                    |
| ACTION  | Yellow warning. Allow creation but suggest scheduling later. |
| RECOVER | Adjust due date or split WO.                                 |

 CAPACITY TIGHT**E-PP03-003****Negative Net Req**

음수 순생산

|         |                                                                          |
|---------|--------------------------------------------------------------------------|
| CAUSE   | FG stock exceeds Order Qty (overstock).                                  |
| DETECT  | Calc check.                                                              |
| ACTION  | Show negative in green (good). Disable WO creation (no need to produce). |
| RECOVER | Skip this row.                                                           |

 SUFFICIENT STOCK**E-PP03-004****BOP Not Defined**

BOP 미정의

|         |                                                       |
|---------|-------------------------------------------------------|
| CAUSE   | MD-04 routing not set for this item.                  |
| DETECT  | JOIN check.                                           |
| ACTION  | Block creation. Cannot determine production sequence. |
| RECOVER | MD-04 BOP entry required.                             |

 NO BOP · SEE MD-04

E-PP03-005

### Concurrent Edit

동시 편집

|         |                                                        |
|---------|--------------------------------------------------------|
| CAUSE   | Same SO selected by another planner.                   |
| DETECT  | Optimistic lock.                                       |
| ACTION  | Block save with current state of conflicting SO shown. |
| RECOVER | Refresh and re-select.                                 |

⚠ CONCURRENT

E-PP03-006

### Past Due Selected

과거 납기 선택

|         |                                                          |
|---------|----------------------------------------------------------|
| CAUSE   | Due date already passed.                                 |
| DETECT  | Date check.                                              |
| ACTION  | Allow with confirmation. WO becomes overdue immediately. |
| RECOVER | Update SO due date in SAP first.                         |

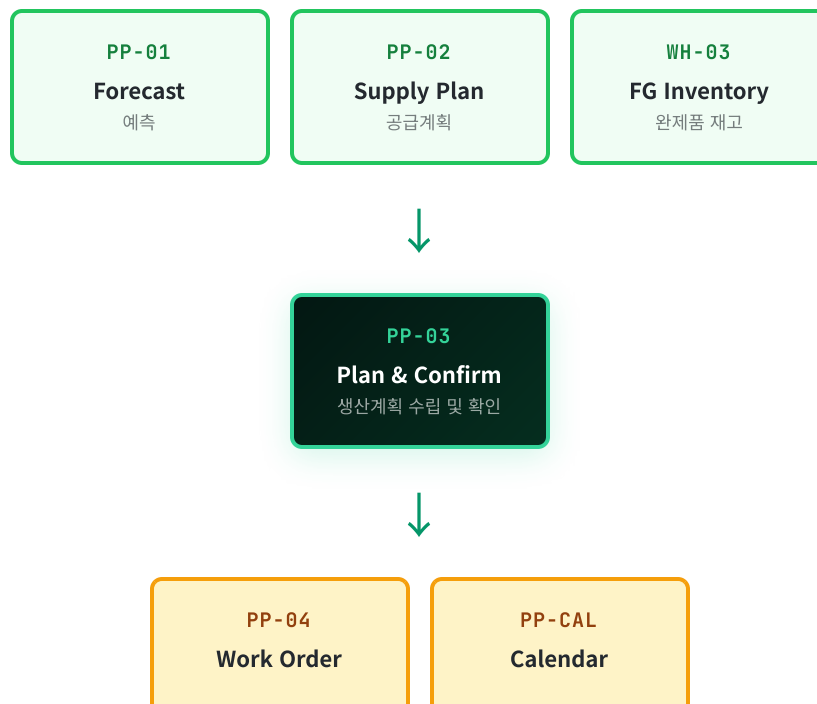
⚠ ALREADY OVERDUE

## 06

## Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



07

Applicable Business Rules

적용 업무 규칙

| RULE ID   | RULE (EN)                                               | 규칙 (KO)                      | ACTION |
|-----------|---------------------------------------------------------|------------------------------|--------|
| BR-PP-001 | Block WO creation if Phase-0 master data is incomplete. | Phase-0 기준정보 미완료 시 WO 생성 차단. | BLK    |
| BR-PP-002 | Auto-block WO when MRP detects material shortage.       | MRP 자재 부족 시 WO 자동 차단.        | BLK    |
| BR-PP-008 | Alert PLAN/MGR when WO due date is within D-3.          | WO 납기 D-3 이내 강조 알림.          | ALT    |

L1A-MES IndexL2VOL.4 · PP Production PlanL3PP-04 · Work Order Management

VOL.4 · PP · PRODUCTION PLANNING · 작업지시 관리

PP-04

WORK ORDER MANAGEMENT

W O M G M T

DEVICEWeb PC

PHASE1st Dev

ROLEPLAN Manager

FREQContinuous · 상시

URL/pp/wo

01

Overview

화면 개요 · 목적 · 컨텍스트

Manages Work Orders through their full lifecycle: Draft → Released → In Progress → Closed. Each WO has a routing badge (A: INJ→IMG→QC→FG · B: INJ→PNT→QC→FG), production progress bar, D-Day indicator, and SAP B1 supply plan reference. Block reasons (Phase-0 incomplete, MRP shortage, equipment unavailable) are surfaced before release.

작업지시(WO) 전체 생명주기 관리: Draft → Released → In Progress → Closed. 각 WO는 라우팅 배지(A: INJ→IMG→QC→FG · B: INJ→PNT→QC→FG), 진척률 바, D-Day, SAP B1 공급계획 참조 표시. 발행 전 차단 사유(Phase-0 미완료·MRP 부족·설비 미가용) 표시.

|            |                                         |
|------------|-----------------------------------------|
| SCREEN ID  | PP-04                                   |
| LIFECYCLE  | Draft → Released → In Progress → Closed |
| ROUTING    | A (INJ→IMG→QC→FG) · B (INJ→PNT→QC→FG)   |
| DOWNSTREAM | INJ/IMG/PNT POP screens                 |
| TABLES     | PP_WO , PP_WOMaterial                   |
| RBAC       | PP.WO                                   |

02

Screen Mockup

화면 목업 — 주요 UI 구성

PP-04 · WORK ORDER MANAGEMENT · WEB PC

A-MES · PP · Work Order Management

PP > WO

All Status▼All Routing▼WO No / Item...

Search

+ Manual WO

12  
Draft

28  
Released

15  
In Progress

3  
Blocked

5  
D-3 Alert

| WO No.          | Item                         | Plan Qty | Routing             | Progress                   | D-Day    | Status       | SAP SO        |
|-----------------|------------------------------|----------|---------------------|----------------------------|----------|--------------|---------------|
| WO-20250801-001 | TRIM-LH-001<br>도어트림 LH       | 2,030    | A:<br>INJ→IMG→QC→FG | <div><div></div></div> 62% | D-12     | In Progress  | SO-2025-08001 |
| WO-20250801-002 | TRIM-LH-001-GY<br>도어트림 LH 회색 | 2,030    | A                   | <div><div></div></div> 30% | D-3<br>⚠ | Released     | SO-2025-08001 |
| WO-20250801-005 | CONS-UP-001<br>콘솔어퍼 (PNT)    | 650      | B:<br>INJ→PNT→QC→FG | —                          | D-5      | MRP Shortage | SO-2025-08003 |

03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트          | TYPE   | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증·동작                                                        | DEFAULT |
|--------|----------------------------|--------|------------|---------------------------------------------------------------------------------------|---------|
| CMP-01 | Status Filter<br>상태 필터     | SELECT | N          | All / Draft / Released / In Progress / Closed / Blocked.                              | All     |
| CMP-02 | Routing Filter<br>라우팅 필터   | SELECT | N          | A: INJ→IMG→QC→FG · B: INJ→PNT→QC→FG.                                                  | All     |
| CMP-03 | Search Box<br>검색           | TEXT   | N          | WO no, item code, or SAP SO no. 300ms debounce.                                       | (empty) |
| CMP-04 | WO Progress Bar<br>진척률 바   | AUTO   | -          | % = (Completed routings / Total routings). Color: green >80%, blue 40-80%, gray <40%. | -       |
| CMP-05 | D-Day Indicator<br>D-Day   | AUTO   | -          | Days = DueDate - TODAY . Red ≤ 3, yellow ≤ 7, green > 7.                              | -       |
| CMP-06 | Manual WO Create<br>수동 WO  | BUTTON | N          | Opens form to create ad-hoc WO not from SAP supply plan. Requires manager role.       | -       |
| CMP-07 | Row Click → Detail<br>행 클릭 | NAV    | -          | Opens WO detail panel with material allocation, routing, history.                     | -       |

04

Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER

page.load · WO list

Loads all WOs with current status, progress, D-Day. Default sort by D-Day ascending.  
전체 WO 목록 로드 (상태·진척률·D-Day). 기본 D-Day 오름차순.

API

GET /api/pp/wo?date=today&sort=dday

2

USER

click · row

Opens detail side panel with WO header, BOM allocation, routing steps, and history log.  
행 클릭 시 상세 패널 표시 (헤더·BOM 할당·라우팅·이력).

API

GET /api/pp/wo/:id/detail



3

**SYSTEM** push · WO status change

When downstream POP (INJ/IMG/PNT) updates routing step, WO progress and status auto-refresh via WebSocket.

POP 모듈(INJ/IMG/PNT) 공정 갱신 시 WebSocket으로 진척률·상태 자동 갱신.

```
PUSH WS event: {wo_id, routing_step, qty}
```

4

**SYSTEM** alert · D-3 approaching

When WO due date is within 3 days, sends alert + highlights row red.

WO 납기 D-3 도달 시 알림 발송 + 행 빨간색 강조.

```
ALERT BR-PP-008 · email + in-app + row.class='rd'
```

5

**USER** click · [+ Manual WO]

Opens manual WO creation form. Requires manager role and provides bypass for non-SAP planning.

수동 WO 생성 폼. 매니저 권한 + SAP 미경유 케이스 처리.

```
API POST /api/pp/wo · source='manual', requires audit reason
```

6

**USER** click · row [Cancel]

User cancels Draft/Released WO. System auto-releases material reservations and reverses status.

Draft/Released WO 취소 시 자재 예약 자동 해제 + 상태 역전.

```
EFFECT BR-PP-004 · WH_Inventory.Reserved 감소
```

7

**SYSTEM** complete · status sync

WO status changes propagate to PP-CAL calendar and PP-OTD KPI calculations.

WO 상태 변경이 PP-CAL 캘린더 및 PP-OTD KPI 집계에 전파.

```
DOWNSTREAM PP-CAL refresh · PP-OTD recalc · RPT-05 dashboard
```

## 05

**Exception Handling**

예외 처리 · 오류 코드 · 복구 절차

**E-PP04-001****MRP Block**

MRP 차단

|         |                                                                      |
|---------|----------------------------------------------------------------------|
| CAUSE   | WO blocked by PP-05 MRP due to material shortage.                    |
| DETECT  | Auto on MRP run.                                                     |
| ACTION  | Row red. Status = Blocked. [Release] disabled. Show shortage detail. |
| RECOVER | Run PP-06 PR or wait for material.                                   |

🚫 BLOCKED · MRP

E-PP04-002

### Cancel In Progress

진행중 취소

|         |                                                                           |
|---------|---------------------------------------------------------------------------|
| CAUSE   | User attempts to cancel In Progress WO.                                   |
| DETECT  | Status check on cancel.                                                   |
| ACTION  | Modal: 'Already in production. Cancel anyway?' Manager approval required. |
| RECOVER | Manager confirms; partial production logged.                              |

⚠️ IN PROGRESS · APPROVE

E-PP04-003

### Routing Mismatch

라우팅 불일치

|         |                                                               |
|---------|---------------------------------------------------------------|
| CAUSE   | POP screen reports routing step inconsistent with BOP.        |
| DETECT  | Push event validation.                                        |
| ACTION  | Yellow warning. Allow but log as anomaly. QC may need review. |
| RECOVER | Audit log entry + QC review.                                  |

⚠️ ROUTING DRIFT

E-PP04-004

### Closed WO Edit Attempt

마감 WO 수정 시도

|         |                                              |
|---------|----------------------------------------------|
| CAUSE   | User tries to modify a Closed WO.            |
| DETECT  | Status check on save.                        |
| ACTION  | Block. WO is immutable after Close.          |
| RECOVER | Create compensating WO if correction needed. |

🚫 CLOSED · IMMUTABLE

E-PP04-005

### Manual WO Permission

수동 WO 권한

|         |                                          |
|---------|------------------------------------------|
| CAUSE   | Non-manager attempts Manual WO.          |
| DETECT  | RBAC check.                              |
| ACTION  | Block with 'PP.WO.Manual role required'. |
| RECOVER | Request role from sys admin.             |

PERMISSION DENIED

E-PP04-006

### WO Qty Mismatch

WO 수량 불일치

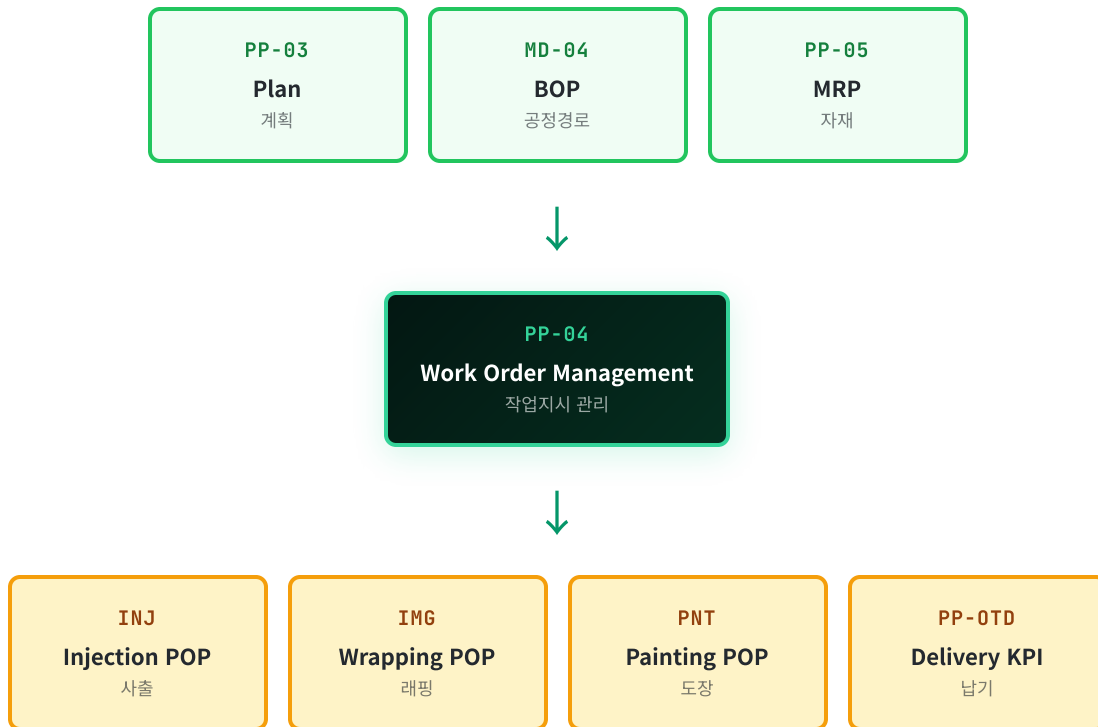
|         |                                                 |
|---------|-------------------------------------------------|
| CAUSE   | Completed qty > planned qty.                    |
| DETECT  | Post-routing validation.                        |
| ACTION  | Yellow flag. Investigate overproduction reason. |
| RECOVER | QC review + reconcile with SO.                  |

△ OVER-PRODUCED

## 06 Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲

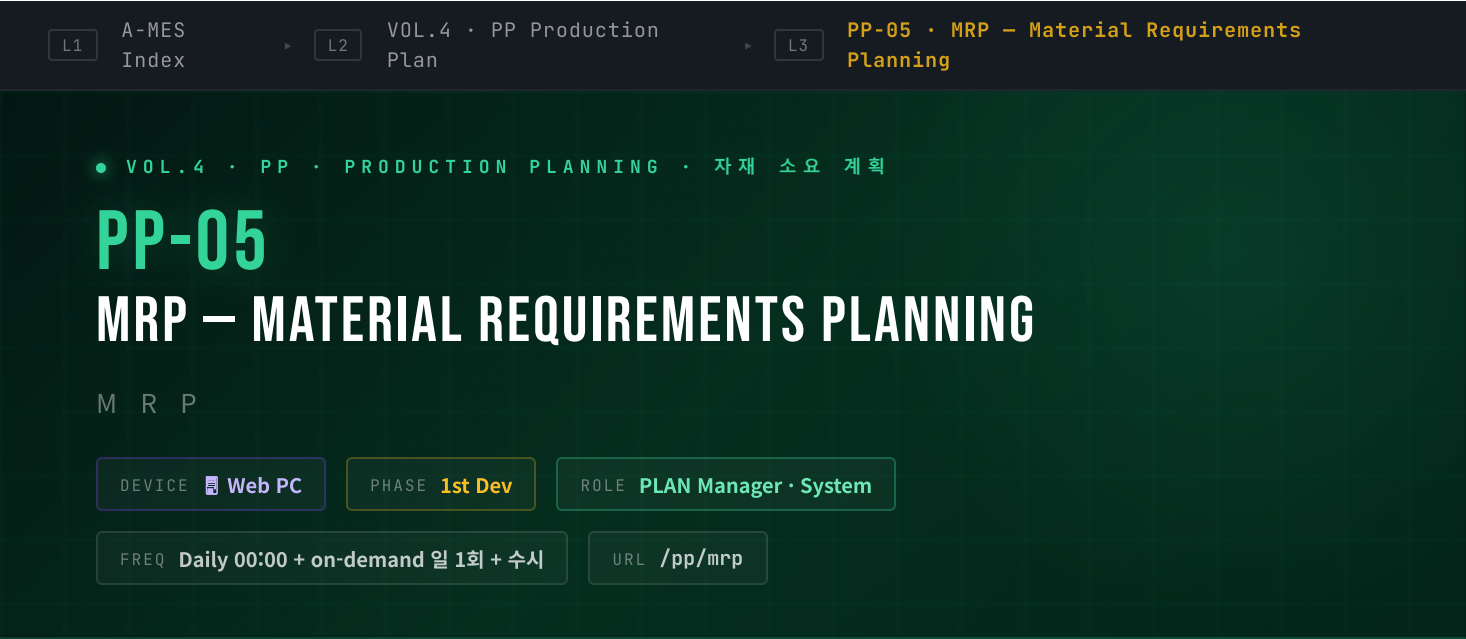


▼ DOWNSTREAM · 하위 화면 / 데이터 ▼

## 07 Applicable Business Rules

적용 업무 규칙

| RULE ID   | RULE (EN)                                         | 규칙 (KO)               | ACTION |
|-----------|---------------------------------------------------|-----------------------|--------|
| BR-PP-002 | Auto-block WO when MRP detects material shortage. | MRP 자재 부족 시 WO 자동 차단. | BLK    |
| BR-PP-003 | Auto-reserve materials on WO release.             | WO 발행 시 자재 자동 예약.     | AUT    |
| BR-PP-004 | Auto-release reserved qty on WO cancellation.     | WO 취소 시 예약 수량 자동 해제.  | AUT    |
| BR-PP-008 | Alert PLAN/MGR when WO due date is within D-3.    | WO 납기 D-3 이내 강조 알림.   | ALT    |



01

Overview

화면 개요 · 목적 · 컨텍스트

Calculates material requirements by exploding BOM for each open WO and comparing against current stock and on-order quantities. Shortage = Required – Stock – OnOrder. Shortages over zero block the affected WOs (IsBlocked=1) and surface for PR generation in PP-06. Auto-runs nightly at 00:00; can be manually triggered for partial recalculation.

각 활성 WO의 BOM 분해 → 현재고·발주중 수량과 대조하여 자재 소요량 계산. 부족량 = 소요 – 재고 – 발주중. 부족 발생 시 해당 WO 차단 (IsBlocked=1)·PP-06 PR 생성 대상 목록 표시. 매일 00:00 자동 실행, 수동 부분 재실행 지원.

|           |                                                              |
|-----------|--------------------------------------------------------------|
| SCREEN ID | PP-05                                                        |
| SCHEDULE  | Daily 00:00 KST + manual                                     |
| FORMULA   | Shortage = Req – Stock – OnOrder                             |
| EFFECT    | WO.IsBlocked=1 on shortage > 0                               |
| TABLES    | PP_WO , MD_BOM , WH_Inventory , WH_PurchaseOrder , PP_MRPLog |
| RBAC      | PP.MRP                                                       |

02

Screen Mockup

화면 목업 — 주요 UI 구성

A-MES · PP · Material Requirements Planning

PP > MRP

Last run: 2025-08-15 00:00:32

▶ Run Now

All WOs

▼

Item...

Export

87

Materials Calc'd

계산 자재

71

Sufficient

충족

8

Shortage

부족

3

WO Blocked

차단 WO

| Material                 | Required<br>소요 | Stock<br>재고 | OnOrder<br>발주중 | Shortage<br>부족 | Lead Time | Order Due<br>발주 기한 | Action  |
|--------------------------|----------------|-------------|----------------|----------------|-----------|--------------------|---------|
| RM-PP-NAT-20<br>PP 펠렛 나뉬 | 3,600 kg       | 2,100 kg    | 500 kg         | -1,000 kg      | 7 days    | 2025-09-08         | + PR 생성 |
| FAB-BK-001<br>원단(흑)      | 1,200 roll     | 800 roll    | 0              | -400 roll      | 14 days   | 2025-09-01         | + PR 생성 |
| ADH-001<br>접착제           | 650 kg         | 980 kg      | 0              | +330 kg        | 5 days    | —                  | —       |
| MAT-INJ-002<br>수지 B      | 2,000 kg       | 2,000 kg    | 0              | ±0             | 10 days   | —                  | —       |

03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트         | TYPE    | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증 · 동작                                              | DEFAULT |
|--------|---------------------------|---------|------------|-------------------------------------------------------------------------------|---------|
| CMP-01 | Last Run Display<br>최근 실행 | DISPLAY | —          | Shows last successful MRP run timestamp with green indicator.                 | —       |
| CMP-02 | [▶ Run Now]<br>수동 실행      | BUTTON  | N          | Triggers immediate MRP run. Locks UI until complete. Manager role required.   | —       |
| CMP-03 | Scope Filter<br>범위 필터     | SELECT  | N          | All WOs / Released only / Blocked only. Affects calculation scope.            | All     |
| CMP-04 | Material Search<br>자재 검색  | TEXT    | N          | Search by item code/name.                                                     | (empty) |
| CMP-05 | Shortage Badge<br>부족 배지   | AUTO    | —          | Red for shortage, gray for balanced, green for surplus.                       | —       |
| CMP-06 | [+ PR 생성]<br>PR 생성        | BUTTON  | N          | Per-row button enabled if shortage. Pre-fills PP-06 form with item, qty, due. | —       |
| CMP-07 | Excel Export<br>엑셀 출력     | BUTTON  | N          | Exports full MRP result table.                                                | —       |

1

**SCHEDULED** cron · 00:00 daily MRP

Auto-runs nightly at 00:00 KST. Iterates all open WOs and explodes BOM via recursive query.

매일 00:00 KST 자동 실행. 활성 WO 전체에 대해 BOM 재귀 분해.

**EFFECT** BR-PP-005 · runtime 5-15 min · result in `PP_MRPLog`

2

**USER** cClick · [▶ Run Now]

Manual MRP trigger. Useful after PP-03 plan changes or PP-02 import.

수동 MRP. PP-03 계획 변경 또는 PP-02 import 후 사용.

**API** `POST /api/pp/mrp/run` · returns job\_id, polling for status

3

**SYSTEM** calc · BOM explosion

For each WO, recursively expand BOM to leaf-level materials. Apply WO.Qty × BOM.ChildQty.

WO별 BOM을 leaf까지 재귀 분해. WO.Qty × BOM.ChildQty로 소요량 누적.

**DB** Recursive CTE on `MD_BOM`

4

**SYSTEM** calc · shortage

Stock = WH\_Inventory.AvailableQty (OnHand - Reserved). OnOrder = WH\_PurchaseOrder.pending. Shortage = Required - Stock - OnOrder.

Stock = WH\_Inventory.AvailableQty (OnHand - Reserved). OnOrder = WH\_PurchaseOrder 미입고. 부족 = 소요 - 재고 - 발주 중.

**OUTPUT** If > 0 → WO.IsBlocked=1 + insert `PP_MRPLog`

5

**SYSTEM** block · affected WOs

WOs with material shortage are auto-blocked. PLAN team notified.

부족 자재 보유 WO 자동 차단. PLAN팀에 알림.

**EFFECT** BR-PP-002 · WO.Status unchanged but IsBlocked=1

6

**USER** cClick · [+ PR 생성]

From shortage row, clicks PR. Pre-fills PP-06 form with item, shortage qty, required date.

부족 행에서 [PR 생성] 클릭 시 PP-06 폼에 자재·부족량·필요일 사전 입력.

**NAV** `/pp/pr?item=...&qty=...&due=...`

7

**SYSTEM** complete · ready for PR

MRP cycle complete. Shortages visible. PR generation ready in PP-06.

MRP 사이클 완료. 부족 자재 표시. PP-06에서 PR 생성 가능.

## 05

## Exception Handling

예외 처리 · 오류 코드 · 복구 절차

**E-PP05-001 BOM Not Found**  
BOM 미정의

|         |                                               |
|---------|-----------------------------------------------|
| CAUSE   | Item in WO has no BOM in MD-02.               |
| DETECT  | BOM lookup.                                   |
| ACTION  | Skip item, log error. WO marked partial calc. |
| RECOVER | MD team defines BOM.                          |

⊘ NO BOM · SEE MD-02

**E-PP05-002 Circular BOM Reference**  
BOM 순환참조

|         |                                  |
|---------|----------------------------------|
| CAUSE   | BOM contains cycle (A→B→A).      |
| DETECT  | Cycle detection.                 |
| ACTION  | Halt calculation, alert MD team. |
| RECOVER | MD team fixes BOM.               |

⊘ CYCLE DETECTED

**E-PP05-003 Long Runtime**  
장시간 실행

|         |                                                       |
|---------|-------------------------------------------------------|
| CAUSE   | MRP run exceeds 15 min.                               |
| DETECT  | Runtime monitor.                                      |
| ACTION  | Continue but flag. Investigate WO count or BOM depth. |
| RECOVER | Optimize: archive Closed WOs.                         |

⚠ SLOW MRP

**E-PP05-004 Database Lock**  
DB 잠금

|       |                                         |
|-------|-----------------------------------------|
| CAUSE | Concurrent inventory update during MRP. |
|-------|-----------------------------------------|



|         |                                    |
|---------|------------------------------------|
| DETECT  | Lock timeout 60s.                  |
| ACTION  | Abort run, retry once. Then alert. |
| RECOVER | Check WH transaction load.         |

DB LOCK · RETRY

#### E-PP05-005 Stale Inventory

재고 데이터 노후화

|         |                                             |
|---------|---------------------------------------------|
| CAUSE   | WH_Inventory.last_updated > 1 hour ago.     |
| DETECT  | Pre-run staleness check.                    |
| ACTION  | Yellow warning. Continue with current data. |
| RECOVER | Check WH module health.                     |

⚠ STALE STOCK

#### E-PP05-006 Massive Shortage

대규모 부족

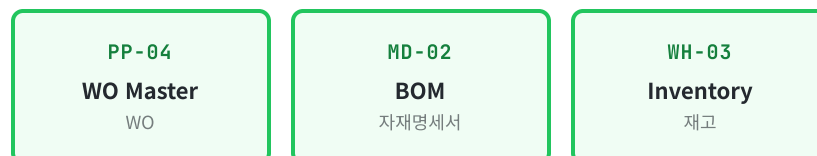
|         |                                                    |
|---------|----------------------------------------------------|
| CAUSE   | Shortage count exceeds 50 materials in single run. |
| DETECT  | Post-run summary.                                  |
| ACTION  | Alert plant director. Likely supply chain issue.   |
| RECOVER | Convene S&OP meeting.                              |

⚠ MASS SHORTAGE

## 06 Linked Screens & Data Flow

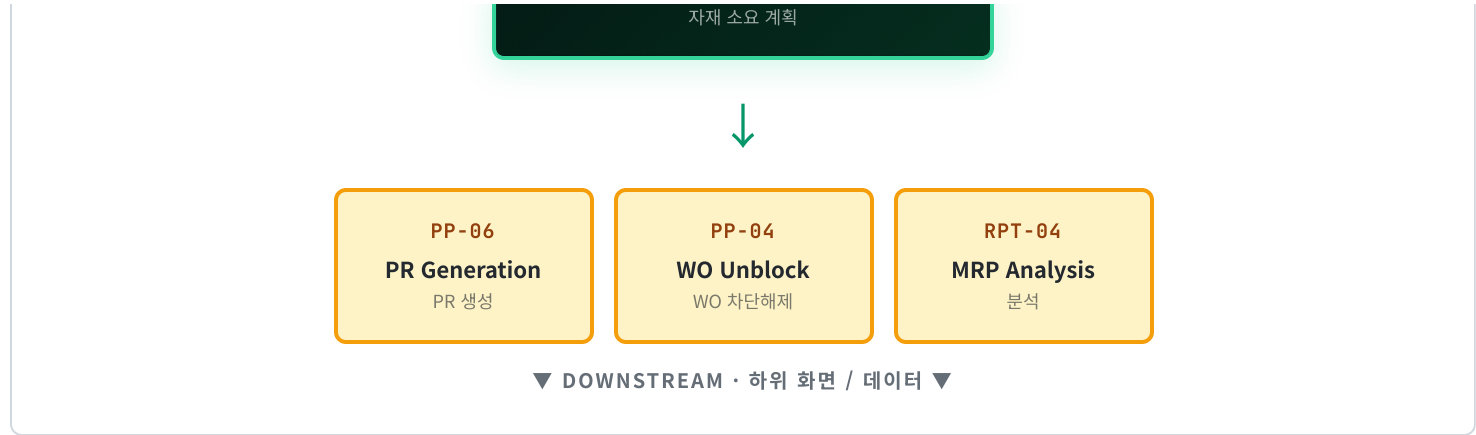
연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



PP-05

MRP — Material Requirements Planning



07

Applicable Business Rules

적용 업무 규칙

| RULE ID   | RULE (EN)                                         | 규칙 (KO)                    | ACTION |
|-----------|---------------------------------------------------|----------------------------|--------|
| BR-PP-002 | Auto-block WO when MRP detects material shortage. | MRP 자재 부족 시 WO 자동 차단.      | BLK    |
| BR-PP-005 | Auto-run MRP nightly at 00:00 KST.                | 매일 00:00 KST MRP 야간 자동 실행. | AUT    |

L1A-MES IndexL2VOL.4 · PP Production PlanL3PP-06 · Purchase Requisition → SAP B1

VOL.4 · PP · PRODUCTION PLANNING · 자재 발주 요청서 (PR) 전송

PP-06PURCHASE REQUISITION → SAP B1

P R S A P B 1

DEVICEWeb PC

PHASE1st Dev

ROLEPLAN Manager

FREQAs needed · 수시

URL/pp/pr

01Overview

화면 개요 · 목적 · 컨텍스트

Generates Purchase Requisitions from MRP shortage results and transmits them to SAP B1 via Service Layer API (POST /PurchaseRequests). Tracks SAP B1 DocNum after successful transmission. Failures auto-retry 3× with backoff; persistent failure escalates to PLAN manager. Approved PRs flow to PO creation in SAP B1 and back to WH-01 (inbound schedule) as expected receipts.

MRP 부족 결과 기반 PR 생성 → SAP B1 Service Layer API(POST /PurchaseRequests) 전송. 성공 시 SAP B1 DocNum 추적. 실패 시 3회 자동 재시도, 지속 실패 시 PLAN 관리자 에스컬레이션. 승인된 PR은 SAP B1에서 PO 생성 → WH-01 입고 예정으로 반영.

|            |                           |
|------------|---------------------------|
| SCREEN ID  | PP-06                     |
| ENDPOINT   | POST /PurchaseRequests    |
| RETRY      | 3× exponential backoff    |
| DOWNSTREAM | SAP B1 PO → WH-01 inbound |
| TABLES     | PP_PR , PP_PRSendLog      |
| RBAC       | PP.PR                     |

02Screen Mockup

화면 목업 — 주요 UI 구성

PP-06 · PURCHASE REQUISITION → SAP B1 · WEB PC

A-MES · PP · Purchase Requisition

PP > PR

All Status

연도 - 월 - 일

PR No / SAP DocNum...

Search

Send to SAP B1

12

Draft  
작성중

8

Sent  
전송완료

42

Approved · PO  
승인·PO 생성

2

Failed  
전송 실패

| PR No.          | SAP DocNum | Item                  | Qty         | Required<br>필요일 | Linked WO           | Status                  | Action  |
|-----------------|------------|-----------------------|-------------|-----------------|---------------------|-------------------------|---------|
| PR-20250801-001 | PR-B1-5542 | RM-PP-NAT-20<br>원자재A  | 1,000<br>kg | 2025-09-08      | WO-20250801-001,002 | Approved · PO           | View PO |
| PR-20250801-002 | PR-B1-5543 | FAB-BK-001<br>원단(흑)   | 400 roll    | 2025-09-01      | WO-20250801-002     | Sent                    | Waiting |
| PR-20250801-003 | —          | RM-MB-GY-03<br>착색 MB  | 50 kg       | 2025-09-12      | WO-20250801-005     | Draft                   | Send    |
| PR-20250730-009 | —          | PNT-BK-NH578<br>블랙 도료 | 100 L       | 2025-08-25      | WO-20250730-007     | Failed · 3/3<br>retries | Retry   |

03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트             | TYPE   | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증·동작                                      | DEFAULT |
|--------|-------------------------------|--------|------------|---------------------------------------------------------------------|---------|
| CMP-01 | Status Filter<br>상태 필터        | SELECT | N          | Draft / Sent / Approved (PO) / Failed.                              | ALL     |
| CMP-02 | Date Filter<br>날짜 필터          | DATE   | N          | Required date for PR.                                               | (empty) |
| CMP-03 | Search<br>검색                  | TEXT   | N          | PR No or SAP DocNum.                                                | (empty) |
| CMP-04 | [ 📄 Send to SAP B1]<br>SAP 전송 | BUTTON | Y          | Sends selected Drafts via Service Layer. Updates status to Sent.    | —       |
| CMP-05 | Retry Counter<br>재시도 카운터      | AUTO   | —          | Display: 'Failed · n/3 retries'. Red after 3 fails.                 | —       |
| CMP-06 | [ 🔄 Retry]<br>재시도             | BUTTON | N          | Per-row retry for failed PRs. Resets retry counter.                 | —       |
| CMP-07 | [View PO]<br>PO 보기            | BUTTON | N          | For approved PRs, deep-link to SAP B1 PO detail (opens in new tab). | —       |

04

Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER

click · select Drafts

User multi-selects Draft PRs from list.  
사용자가 Draft PR 다중 선택.

UI

Checkbox column · footer shows selected count

2

USER

click · [ 📄 Send]

Triggers transmission. Each PR sent individually via API to SAP B1 Service Layer.  
전송 트리거. 각 PR이 SAP B1 Service Layer API로 개별 전송.

API

POST /b1s/v1/PurchaseRequests per row

3

SYSTEM

transmit · retry on fail

On failure, auto-retries 3× with backoff (5s, 15s, 45s). After 3 failures, marks Failed.  
실패 시 지수 backoff(5s/15s/45s)로 3회 자동 재시도. 3회 실패 후 Failed 상태.

EFFECT

BR-PP-006 · log all attempts in PP\_PRSendLog

4

SYSTEM

track · DocNum

On success, SAP B1 returns DocNum. Stored in PP\_PR.SAP\_DocNum. Status = Sent.

성공 시 SAP B1이 DocNum 반환. PP\_PR.SAP\_DocNum에 저장 + 상태 Sent.

DB Status: Draft → Sent

5

SCHEDULED poll · PO creation

Every 1 hour, polls SAP B1 to check if any Sent PR has been converted to PO.

1시간마다 SAP B1 폴링하여 Sent PR의 PO 전환 여부 확인.

API GET /b1s/v1/PurchaseOrders?\$filter=PurchaseRequestNum eq ...

6

SYSTEM update · PO Created

When SAP B1 generates PO from PR, status → Approved (PO). Triggers WH-01 inbound schedule update.

SAP B1에서 PR이 PO로 전환되면 상태 Approved (PO) → WH-01 입고 예정 갱신.

DOWNSTREAM WH\_PurchaseOrder INSERT + WH-01 KPI refresh

7

USER click · [🔄 Retry]

Manual retry for persistently failed PRs (after fixing root cause).

지속 실패 PR에 대한 수동 재시도 (근본 원인 해결 후).

RESET Retry counter reset, attempt fresh transmission

## 05 Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PP06-001

### SAP Auth Failure

SAP 인증 실패

CAUSE Service Layer token expired or invalid.

DETECT 401 response.

ACTION Auto re-login. Retry. If still fails, alert SYS.

RECOVER SYS updates credential.

🚫 AUTH FAILED

E-PP06-002

### PR Validation Reject

PR 검증 거부

CAUSE SAP B1 rejects PR (invalid vendor, item, etc.).

DETECT 400 response with reason.

ACTION Mark Failed with SAP error message. Skip retries.

RECOVER User fixes data in PP-06 or MD masters.

🚫 SAP REJECTED

E-PP06-003

### Network Timeout

네트워크 타임아웃

|         |                               |
|---------|-------------------------------|
| CAUSE   | Service Layer response > 30s. |
| DETECT  | Heartbeat.                    |
| ACTION  | Retry 3× per BR-PP-006.       |
| RECOVER | Check SAP B1 health.          |

⚠️ TIMEOUT · RETRYING

E-PP06-004

### Duplicate PR

PR 중복

|         |                                                 |
|---------|-------------------------------------------------|
| CAUSE   | Same item+date already has open PR.             |
| DETECT  | Pre-send check.                                 |
| ACTION  | Show duplicate, ask user: send anyway or merge. |
| RECOVER | Merge or cancel duplicate.                      |

⚠️ DUPLICATE

E-PP06-005

### Insufficient Vendor

거래선 부재

|         |                                              |
|---------|----------------------------------------------|
| CAUSE   | Item has no default vendor in MD-07.         |
| DETECT  | Pre-send.                                    |
| ACTION  | Block send. Cannot create PR without vendor. |
| RECOVER | MD team sets default vendor in MD-07.        |

🚫 NO VENDOR

E-PP06-006

### SAP B1 Down

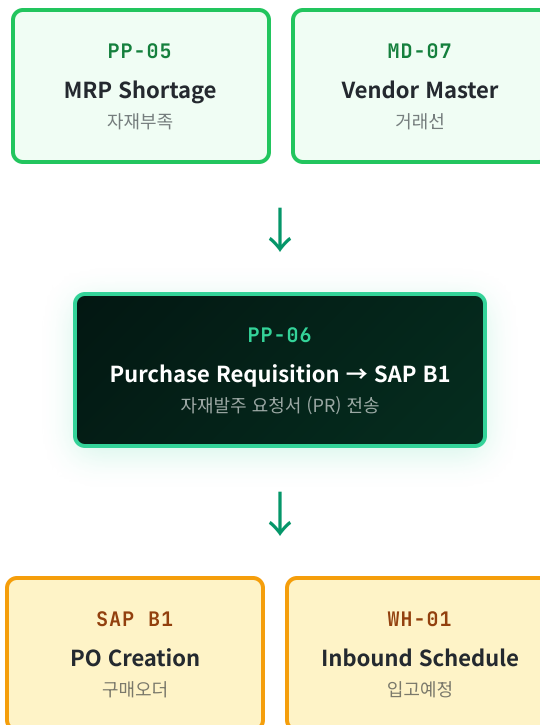
SAP 다운

|         |                                               |
|---------|-----------------------------------------------|
| CAUSE   | Repeated 5xx errors from SAP.                 |
| DETECT  | Status check.                                 |
| ACTION  | Halt all sends. Banner: 'SAP B1 unavailable'. |
| RECOVER | Wait for SAP recovery, then resume.           |

## 06 Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



▼ DOWNSTREAM · 하위 화면 / 데이터 ▼

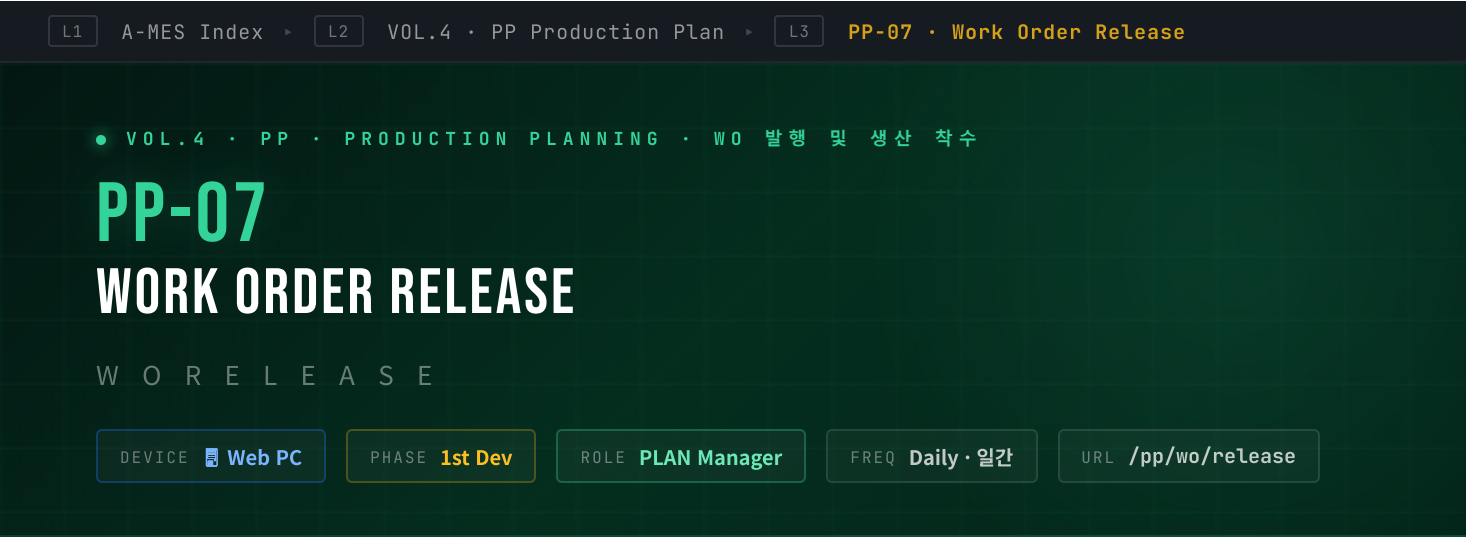
## 07 Applicable Business Rules

적용 업무 규칙

| RULE ID | RULE (EN) | 규칙 (KO) | ACTION |
|---------|-----------|---------|--------|
|---------|-----------|---------|--------|

|           |                                                             |                                         |     |
|-----------|-------------------------------------------------------------|-----------------------------------------|-----|
| BR-PP-006 | Retry SAP B1 PR transmission 3× on failure; escalate after. | SAP B1 PR 전송 실패 시 3회 재시도, 지속 시 PLAN 알림. | AUT |
|-----------|-------------------------------------------------------------|-----------------------------------------|-----|





01

## Overview

화면 개요 · 목적 · 컨텍스트

Validates all 4 release conditions sequentially before transitioning WO to Released status: **(1) Phase-0 complete, (2) MRP no shortage, (3) Equipment available, (4) Delivery lead time sufficient.** On release, materials are auto-reserved (WH\_Inventory.Reserved += qty) and the WO becomes visible on INJ/IMG/PNT POP screens. Cancellation auto-releases reservations.

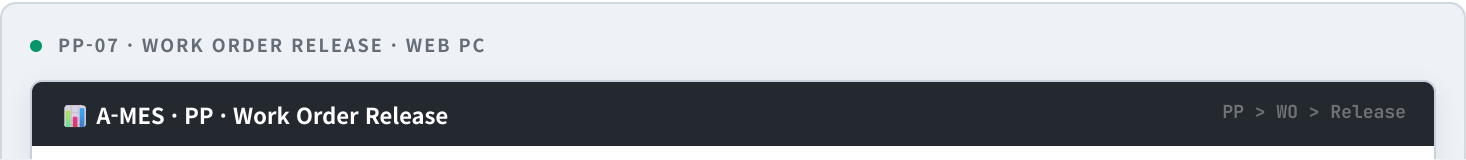
WO Released 전환 전 4가지 조건 순차 검증: **(1) Phase-0 완료 · (2) MRP 자재 충족 · (3) 설비 가용 · (4) 납기 여유.** 발행 시 자재 자동 예약 (WH\_Inventory.Reserved += qty), INJ/IMG/PNT POP 화면 표시. 취소 시 예약 자동 해제.

|           |                                       |
|-----------|---------------------------------------|
| SCREEN ID | PP-07                                 |
| GATES     | 4 sequential validations              |
| EFFECT    | Material reservation + POP visibility |
| CANCEL    | Auto-release reservation (BR-PP-004)  |
| TABLES    | PP_WO , WH_Inventory                  |
| RBAC      | PP.WO.Release                         |

02

## Screen Mockup

화면 목업 — 주요 UI 구성



▶ WO-20250801-001 · TRIM-LH-001 · 2,030 EA · 발행 검증

GATE 1

✓ Phase-0 Complete

기준정보 완료

BR-PP-001 OK

GATE 2

✓ MRP Sufficient

자재 충족

BR-PP-002 OK

GATE 3

✓ Equipment Available

설비 가용

INJ-L1 OK

GATE 4

⚠ Tight (D-12)

납기 여유

Above min 7d

✓ Release WO

📄 Save Draft

자재 자동 예약: RM-PP-NAT-20 × 81 kg + FAB-GY-C03 × 12 roll

| WO No.          | Item        | Phase-0 | MRP | Equipment | Lead Time | Result            |
|-----------------|-------------|---------|-----|-----------|-----------|-------------------|
| WO-20250801-001 | TRIM-LH-001 | ✓       | ✓   | ✓         | ⚠ D-12    | Ready             |
| WO-20250801-005 | CONS-UP-001 | ✓       | ✗   | ✓         | ✓         | Blocked · MRP     |
| WO-20250801-007 | TRIM-RH-002 | ✗       | —   | —         | —         | Blocked · Phase-0 |

## 03 UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트              | TYPE    | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증 · 동작                                                  | DEFAULT |
|--------|--------------------------------|---------|------------|-----------------------------------------------------------------------------------|---------|
| CMP-01 | Gate 1 · Phase-0<br>기준정보       | AUTO    | Y          | Checks BOM, BOP, Inspection Standard exist. BR-PP-001 .                           | —       |
| CMP-02 | Gate 2 · MRP<br>MRP 충족         | AUTO    | Y          | Checks WO.IsBlocked == 0 from latest MRP run.                                     | —       |
| CMP-03 | Gate 3 · Equipment<br>설비 가용    | AUTO    | Y          | Checks MNT_Equipment.Status ≠ 'Blocked' for routing equipment.                    | —       |
| CMP-04 | Gate 4 · Lead Time<br>납기 여유    | AUTO    | Y          | Days until due ≥ min lead time per BOP . Min 7 days default.                      | —       |
| CMP-05 | [ ✓ Release WO ]<br>WO 발행      | BUTTON  | Y          | Enabled only when all 4 gates pass. Updates Status=Released + reserves materials. | —       |
| CMP-06 | [ 📄 Save Draft ]<br>Draft 저장   | BUTTON  | N          | Keeps WO in Draft for later release.                                              | —       |
| CMP-07 | Reservation Preview<br>예약 미리보기 | DISPLAY | —          | Shows materials and qty that will be reserved on release.                         | —       |

1

**USER** page.load · select WO

User opens release screen for a specific Draft WO. System runs all 4 gate checks in parallel.

Draft WO 발행 화면 진입. 4개 게이트를 병렬 검증.

**API** GET /api/pp/wo/:id/release-check

2

**SYSTEM** validate · Gate 1 Phase-0

Joins WO.item\_code with MD-02 (BOM), MD-04 (BOP), MD-06 (Inspection). All must exist.

MD-02/04/06 존재 여부 검증.

**RESULT** PASS → green check · FAIL → block with specific missing master

3

**SYSTEM** validate · Gate 2 MRP

Checks WO.IsBlocked set by last MRP. Also re-runs partial MRP for this WO if &gt; 24h since last full.

MRP 실행 후 WO.IsBlocked 확인 + 24시간 경과 시 부분 재실행.

**EFFECT** BR-PP-002 · auto-trigger partial MRP if stale

4

**SYSTEM** validate · Gate 3 Equipment

For each routing step in BOP, checks MNT\_Equipment availability for the expected start period.

BOP 라우팅별 시작 시점의 설비 가용성 확인.

**QUERY** SELECT status FROM mnt\_equipment WHERE line\_id IN (...) AND start\_time BETWEEN ...

5

**SYSTEM** validate · Gate 4 Lead

DueDate - TODAY ≥ MinLeadDays (from BOP). Default 7 days.

납기 - 오늘 ≥ 최소 리드(BOP 정의). 기본 7일.

**RESULT** PASS (green) / WARN (≥3, <7, yellow) / FAIL (<3, red)

6

**USER** click · [Release WO]

All gates pass (or user accepts warnings). Transitions Status to Released.

전체 게이트 통과 (또는 경고 수락) 시 Status를 Released로 전환.

**API** POST /api/pp/wo/:id/release

7

**SYSTEM** complete · reserve + visible

Materials auto-reserved in WH\_Inventory. WO becomes visible on INJ/IMG/PNT POP. Calendar updates.

자재 자동 예약 + INJ/IMG/PNT POP 표시. 캘린더 갱신.

**DOWNSTREAM** BR-PP-003 · WH.Reserved += qty · POP visibility · PP-CAL refresh

## 05 Exception Handling

예외 처리 · 오류 코드 · 복구 절차

### E-PP07-001 Gate 1 Failure

기준정보 미완료

|         |                                              |
|---------|----------------------------------------------|
| CAUSE   | Phase-0 incomplete.                          |
| DETECT  | Gate check.                                  |
| ACTION  | Block release. Show which master is missing. |
| RECOVER | MD team completes.                           |

🚫 GATE 1 · PHASE-0

### E-PP07-002 Gate 2 Failure

MRP 차단

|         |                                |
|---------|--------------------------------|
| CAUSE   | Material shortage from MRP.    |
| DETECT  | Gate check.                    |
| ACTION  | Block. Link to PP-06 for PR.   |
| RECOVER | Generate PR, wait for inbound. |

🚫 GATE 2 · MRP

### E-PP07-003 Gate 3 Warning

설비 경고

|         |                                                       |
|---------|-------------------------------------------------------|
| CAUSE   | Equipment scheduled for maintenance during WO period. |
| DETECT  | Gate check.                                           |
| ACTION  | Warning popup. Allow with manager approval.           |
| RECOVER | Approve and adjust schedule, or postpone.             |

⚠️ GATE 3 · MNT

### E-PP07-004 Gate 4 Tight

납기 압박

|         |                                       |
|---------|---------------------------------------|
| CAUSE   | Lead time < 7 days.                   |
| DETECT  | Gate check.                           |
| ACTION  | Warning. Release allowed but flagged. |
| RECOVER | Expedite or adjust due date.          |

E-PP07-005

### Reservation Conflict

예약 충돌

|         |                                                |
|---------|------------------------------------------------|
| CAUSE   | Material already fully reserved by other WOs.  |
| DETECT  | Reservation transaction.                       |
| ACTION  | Block. Reservation cannot exceed AvailableQty. |
| RECOVER | Wait for inbound or cancel other WO.           |

⊖ RESERVATION

E-PP07-006

### Release Permission

발행 권한 없음

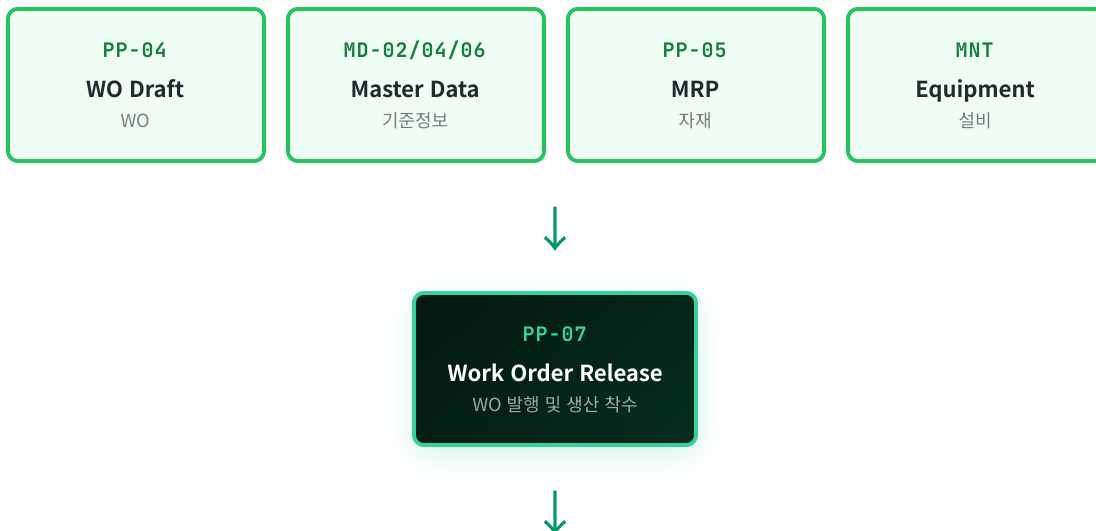
|         |                                   |
|---------|-----------------------------------|
| CAUSE   | User lacks PP.WO.Release role.    |
| DETECT  | RBAC.                             |
| ACTION  | Block. Only managers can release. |
| RECOVER | Manager releases or assigns role. |

⊖ PERMISSION

## 06 Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



INJ  
Injection POP  
사출

IMG/PNT  
Wrapping/Painting POP  
래핑·도장

PP-CAL  
Calendar  
캘린더

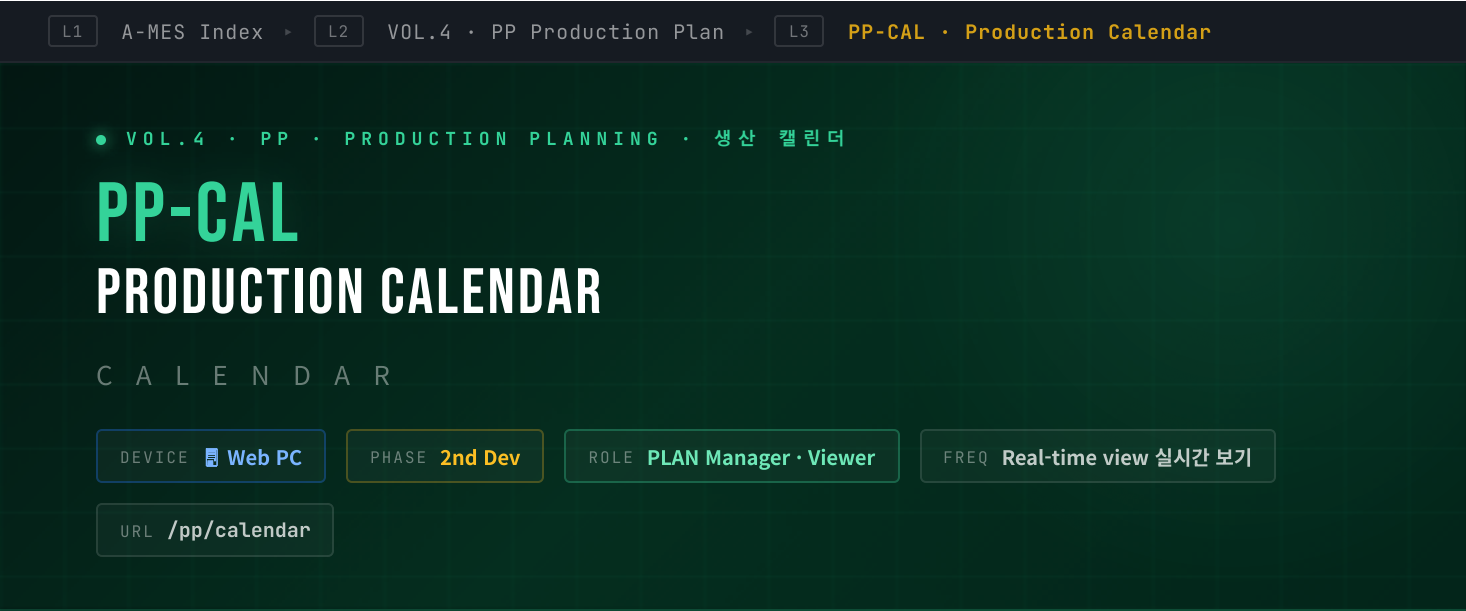
▼ DOWNSTREAM · 하위 화면 / 데이터 ▼

07

Applicable Business Rules

적용 업무 규칙

| RULE ID   | RULE (EN)                                               | 규칙 (KO)                      | ACTION |
|-----------|---------------------------------------------------------|------------------------------|--------|
| BR-PP-001 | Block WO creation if Phase-0 master data is incomplete. | Phase-0 기준정보 미완료 시 WO 생성 차단. | BLK    |
| BR-PP-002 | Auto-block WO when MRP detects material shortage.       | MRP 자재 부족 시 WO 자동 차단.        | BLK    |
| BR-PP-003 | Auto-reserve materials on WO release.                   | WO 발행 시 자재 자동 예약.            | AUT    |
| BR-PP-007 | Warn if equipment is blocked when releasing WO.         | WO 발행 시 설비 차단 상태이면 경고.       | ALT    |



01

Overview

화면 개요 · 목적 · 컨텍스트

Calendar view of WO due dates with line capacity overlay. Each day shows scheduled WO count, total qty, and capacity utilization. D-3 critical items highlighted red, due-today in orange. Drag-drop to reschedule WO due date (with audit log + manager approval if past today).

WO 납기일을 캘린더 뷰로 표시. 일별로 예정 WO 수·총 수량·능력 가동률 표시. D-3 이내 빨강, 금일 납기 주황 강조. 드래그-드롭으로 납기 재조정(감사 로그 + 과거 일자는 매니저 승인).

|             |                                   |
|-------------|-----------------------------------|
| SCREEN ID   | PP-CAL                            |
| VIEW        | Month / Week / Day                |
| D-DAY ALERT | D-3 red, today orange (BR-PP-008) |
| RESCHEDULE  | Drag-drop + audit log             |
| TABLES      | PP_WO , MD_Equipment (capacity)   |
| RBAC        | PP.WO                             |

02

Screen Mockup

화면 목업 — 주요 UI 구성

&lt; Aug

September 2025

Oct &gt;

Month View ▼

All Lines ▼

Today

| Sun                         | Mon | Tue | Wed                | Thu | Fri | Sat |
|-----------------------------|-----|-----|--------------------|-----|-----|-----|
| 1                           | 2   | 3   | 4                  | 5   | 6   | 7   |
| 8                           | 9   | 10  | 11                 | 12  | 13  | 14  |
| 15<br>3 WO · 4,820<br>△ D-3 | 16  | 17  | 18<br>2 WO · 1,580 | 19  | 20  | 21  |
| 22                          | 23  | 24  | 25<br>1 WO · 800   | 26  | 27  | 28  |
| 29                          | 30  |     |                    |     |     |     |

■ D-3 Critical · 납기 임박   ■ Approaching · 접근중   ■ Normal · 정상

## 03

## UI Components Specification

UI 컴포넌트 명세 · 검증 규칙



| ID     | COMPONENT<br>컴포넌트            | TYPE     | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증·동작                                                    | DEFAULT |
|--------|------------------------------|----------|------------|-----------------------------------------------------------------------------------|---------|
| CMP-01 | View Mode<br>보기 모드           | SELECT   | N          | Month / Week / Day. Affects grid density.                                         | Month   |
| CMP-02 | Line Filter<br>라인 필터         | SELECT   | N          | Filter calendar by production line.                                               | ALL     |
| CMP-03 | Today Button<br>오늘           | BUTTON   | N          | Jumps to today's date.                                                            | —       |
| CMP-04 | Day Cell<br>일자 셀             | INTERACT | —          | Shows WO count, total qty, capacity %. Click expands list. Drag-drop reschedules. | —       |
| CMP-05 | WO Drag-Drop<br>WO 드래그       | INTERACT | N          | Drag WO chip to another day to reschedule. Past date requires manager approval.   | —       |
| CMP-06 | D-Day Color Code<br>D-Day 색상 | AUTO     | —          | Red ≤ D-3, orange today, green normal. BR-PP-008 .                                | —       |
| CMP-07 | Capacity Overlay<br>능력 오버레이  | DISPLAY  | —          | Per-day load %. Red > 90%, yellow 70-90%.                                         | —       |

04

Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER

page.load · open calendar

Renders current month with all WOs by due date. Capacity overlay loaded from MNT.  
현재 월에 WO 표시 + MNT 능력 오버레이.

API

GET /api/pp/calendar?month=YYYY-MM

2

USER

click · day cell

Expands the day to show full WO list with item, qty, status.  
일자 셀 클릭 시 전체 WO 목록(품목·수량·상태) 표시.

UI

Modal or expand-row

3

USER

drag · WO to new date

User drags WO chip to another day. System validates: not past today, capacity available, lead time OK.  
WO를 다른 일자로 드래그. 과거 일자·능력·리드타임 검증.

VALIDATE

Block past date without approval. Capacity check.

SYSTEM

save · reschedule

4

Updates PP\_WO.due\_date with audit log entry.

PP\_WO.due\_date 갱신 + 감사 로그.

**AUDIT** Audit: old\_date, new\_date, user, reason

5

**SYSTEM** alert · D-3 daily check

Every day at 00:00, scans all WOs and flags those reaching D-3 today.

매일 00:00 D-3 도달 WO 자동 검색·플래그.

**ALERT** BR-PP-008 · email + UI highlight

6

**SYSTEM** sync · downstream

Reschedule propagates to PP-04 list and PP-OTD KPI projection.

재조정이 PP-04 목록·PP-OTD KPI 예측에 반영.

**DOWNSTREAM** PP-04 / PP-OTD refresh

## 05 Exception Handling

예외 처리 · 오류 코드 · 복구 절차

E-PPCAL-001

### Past Date Drop

과거 일자 드롭

|         |                                                                |
|---------|----------------------------------------------------------------|
| CAUSE   | User drops WO to a past date.                                  |
| DETECT  | Drop validation.                                               |
| ACTION  | Block. Modal: 'Cannot reschedule to past. Approve as manager?' |
| RECOVER | Manager override with reason.                                  |

⚠ PAST DATE

E-PPCAL-002

### Capacity Exceeded

능력 초과

|         |                                  |
|---------|----------------------------------|
| CAUSE   | New date capacity > 100%.        |
| DETECT  | Capacity check.                  |
| ACTION  | Yellow warning. Allow with note. |
| RECOVER | Confirm with capacity manager.   |

⚠ OVER CAPACITY

### Lead Time Violation

#### E-PPCAL-003 리드타임 위반

|         |                                                  |
|---------|--------------------------------------------------|
| CAUSE   | New date doesn't allow required production lead. |
| DETECT  | Lead check.                                      |
| ACTION  | Block. Show min required date.                   |
| RECOVER | Choose later date.                               |

LEAD INSUFFICIENT

#### E-PPCAL-004 Closed WO Drag 마감 WO 드래그

|         |                                  |
|---------|----------------------------------|
| CAUSE   | User drags a Closed WO.          |
| DETECT  | Status check.                    |
| ACTION  | Block. Closed WOs are immutable. |
| RECOVER | —                                |

CLOSED

#### E-PPCAL-005 Calendar Load Slow 캘린더 로딩 지연

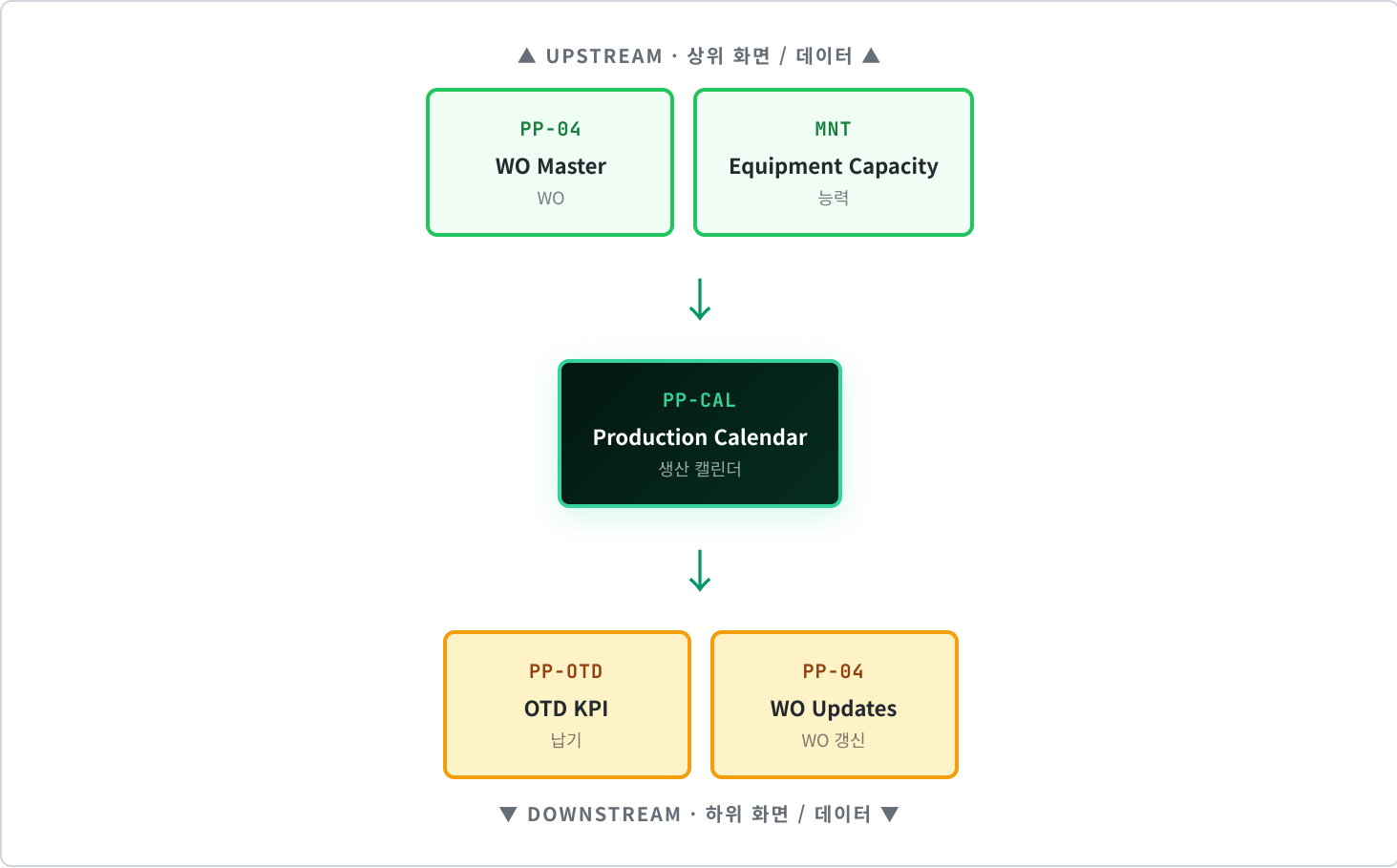
|         |                                         |
|---------|-----------------------------------------|
| CAUSE   | Many WOs in month, render slow.         |
| DETECT  | Render time monitor.                    |
| ACTION  | Show loading spinner. Virtualize cells. |
| RECOVER | Narrow filter.                          |

SLOW LOAD

#### E-PPCAL-006 Permission to Reschedule 재조정 권한

|         |                                |
|---------|--------------------------------|
| CAUSE   | User lacks PP.WO.Edit role.    |
| DETECT  | RBAC.                          |
| ACTION  | Drag disabled. View-only mode. |
| RECOVER | Request role.                  |

READ ONLY



07Applicable Business Rules

적용 업무 규칙

| RULE ID   | RULE (EN)                                      | 규칙 (KO)             | ACTION |
|-----------|------------------------------------------------|---------------------|--------|
| BR-PP-008 | Alert PLAN/MGR when WO due date is within D-3. | WO 납기 D-3 이내 강조 알림. | ALT    |



## 01 Overview

화면 개요 · 목적 · 컨텍스트

Tracks On-Time Delivery (OTD) KPI by customer (SAV/GEO), item, and period. Calculates  $OTD\% = (\text{Shipped on or before due} / \text{Total shipped}) \times 100$ . Targets: SAV  $\geq 98\%$ , GEO  $\geq 96\%$ . Shows trend chart (12-month), delay root-cause Pareto, and individual delayed shipment drill-down. Feeds RPT-05 monthly KPI dashboard.

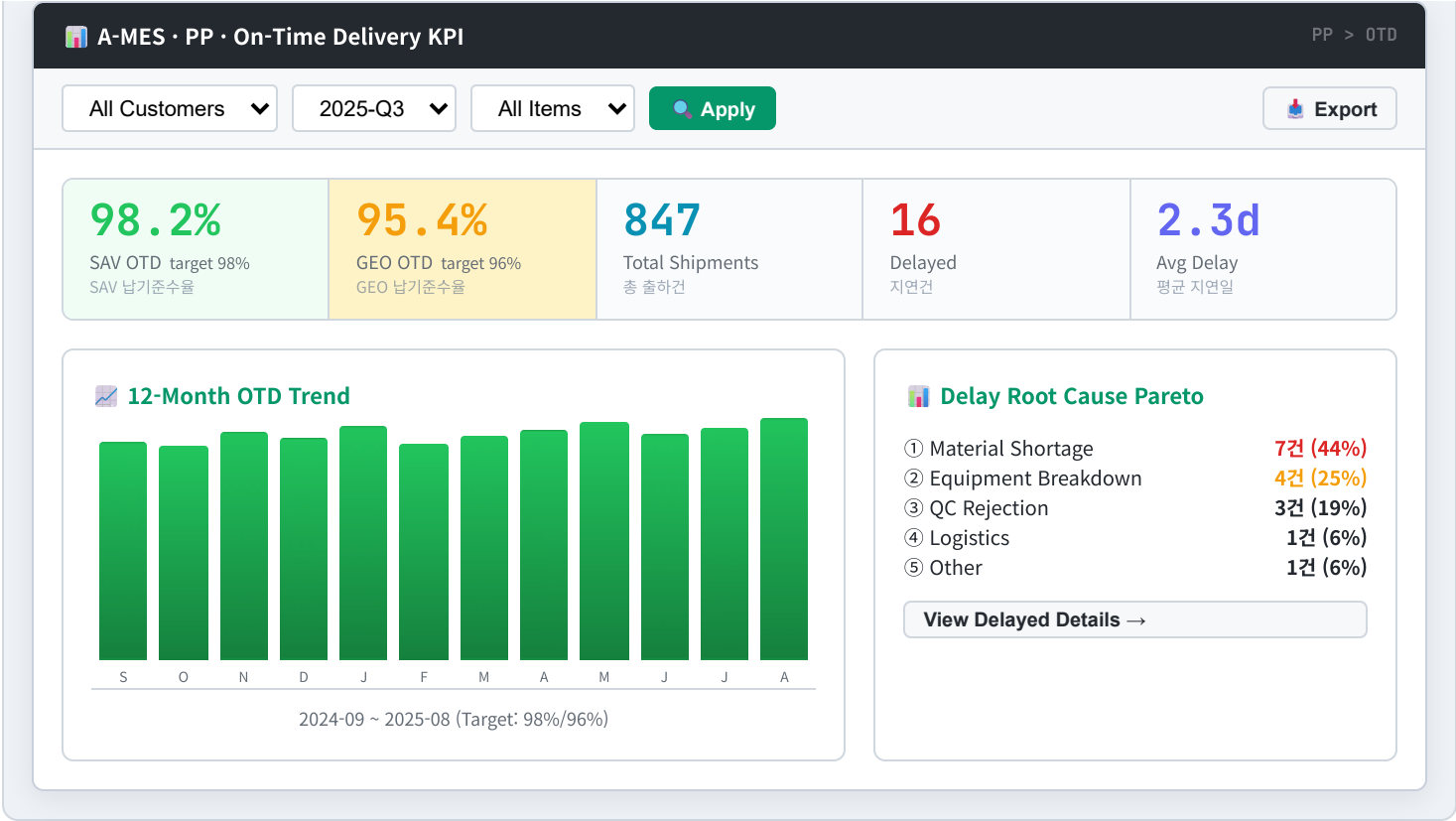
거래선(SAV/GEO)·품목·기간별 OTD KPI 추적.  $OTD\% = (\text{납기 내 출하} / \text{총 출하}) \times 100$ . 목표: SAV  $\geq 98\%$  · GEO  $\geq 96\%$ . 12개월 추이 차트·지연 원인 Pareto·개별 지연건 드릴다운. RPT-05 월간 KPI 대시보드 연계.

|           |                                           |
|-----------|-------------------------------------------|
| SCREEN ID | PP-OTD                                    |
| TARGET    | SAV $\geq 98\%$ · GEO $\geq 96\%$         |
| FORMULA   | (On-time / Total) $\times 100$            |
| SOURCE    | FG_Shipment.actual_date vs PP_WO.due_date |
| TABLES    | FG_Shipment , PP_WO                       |
| RBAC      | PP.View · Director                        |

## 02 Screen Mockup

화면 목업 — 주요 UI 구성





03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트               | TYPE    | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증 · 동작                              | DEFAULT   |
|--------|---------------------------------|---------|------------|---------------------------------------------------------------|-----------|
| CMP-01 | Customer Filter<br>고객사          | SELECT  | N          | All / SAV / GEO.                                              | All       |
| CMP-02 | Period Filter<br>기간             | SELECT  | N          | Q/M/Custom date range.                                        | Current Q |
| CMP-03 | Item Filter<br>품목               | SELECT  | N          | Filter by specific item.                                      | All       |
| CMP-04 | OTD Card<br>OTD 카드              | DISPLAY | -          | <div>OTD% = (On-time / Total) × 100</div> . Target indicator. | -         |
| CMP-05 | Trend Chart<br>추이 차트            | CHART   | -          | 12-month bar chart. Click bar drills into that month.         | -         |
| CMP-06 | Pareto Chart<br>Pareto          | CHART   | -          | Top 5 delay reasons. Click row drills to detailed cases.      | -         |
| CMP-07 | [View Delayed Details]<br>지연 상세 | BUTTON  | N          | Drill into individual delayed shipments with root cause.      | -         |

## 04 Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

**USER** page.load · open OTD

Loads current period OTD with all filters at default. Calls aggregate API.

현재 기간 OTD를 기본 필터로 로드. 집계 API 호출.

**API** GET /api/pp/otd?period=current\_quarter

2

**SYSTEM** calc · OTD%

Aggregates: FG\_Shipment joined with PP\_WO. Counts where actual\_ship\_date ≤ wo\_due\_date.

FG\_Shipment × PP\_WO 조인. actual\_ship\_date ≤ wo\_due\_date 건수 집계.

**FORMULA** OTD% = COUNT(on\_time) / COUNT(\*) \* 100

3

**USER** click · trend bar

Drills into selected month. Re-fetches detailed shipments for that month.

추이 차트 막대 클릭 시 해당 월 상세 출하 데이터 조회.

**API** GET /api/pp/otd/detail?month=YYYY-MM

4

**USER** click · Pareto row

Drills into shipments with that delay reason. Shows root cause analysis link to QC/MNT data.

Pareto 행 클릭 시 해당 사유 지연건 목록 + QC/MNT 데이터 링크.

**NAV** /pp/otd/detail?cause=MaterialShortage

5

**SCHEDULED** cron · daily KPI snapshot

Every day at 00:30, snapshots OTD for previous day into PP\_OTD\_Snapshot.

매일 00:30 전일 OTD 스냅샷을 PP\_OTD\_Snapshot에 저장.

**EFFECT** Used for fast trend rendering

6

**SYSTEM** feed · monthly KPI

Aggregated to RPT-05 monthly KPI dashboard. Director-level visibility.

월간 RPT-05 KPI 대시보드에 집계 전달. 임원진 가시성.

**DOWNSTREAM** RPT-05 · Email digest to plant director on miss

## 05 Exception Handling

예외 처리 · 오류 코드 · 복구 절차

**E-PP0TD-001****No Shipments**

출하 없음

CAUSE No FG shipments in filter range.

DETECT Empty data.

ACTION Show empty state.

RECOVER Widen date range.

i NO DATA

**E-PP0TD-002****Below Target**

목표 미달

CAUSE OTD &lt; target (SAV 98% / GEO 96%).

DETECT Daily check.

ACTION Red KPI card. Alert plant director.

RECOVER Root cause analysis + CAPA.

⚠ BELOW TARGET

**E-PP0TD-003****Missing WO Reference**

WO 참조 누락

CAUSE Shipment has no linked WO (manual ship?).

DETECT Join NULL.

ACTION Excluded from OTD calc. Listed separately.

RECOVER Reconcile in FG module.

⚠ ORPHAN SHIPMENT

**E-PP0TD-004****Slow Query**

쿼리 지연

CAUSE Aggregation &gt; 10s for large period.

DETECT Timeout monitor.

ACTION Use snapshot table or narrow scope.

RECOVER Apply more specific filter.

⚠ SLOW · USE SNAPSHOT



E-PP0TD-005

### Snapshot Missing

스냅샷 결손

|         |                                                 |
|---------|-------------------------------------------------|
| CAUSE   | Daily snapshot job failed yesterday.            |
| DETECT  | Existence check.                                |
| ACTION  | Re-aggregate from raw. Mark with warning badge. |
| RECOVER | SYS reruns snapshot job.                        |

⚠ NO SNAPSHOT

E-PP0TD-006

### Export Permission

출력 권한 없음

|         |                                |
|---------|--------------------------------|
| CAUSE   | User lacks PP.OTD.Export role. |
| DETECT  | RBAC.                          |
| ACTION  | Block export. View only.       |
| RECOVER | Request role.                  |

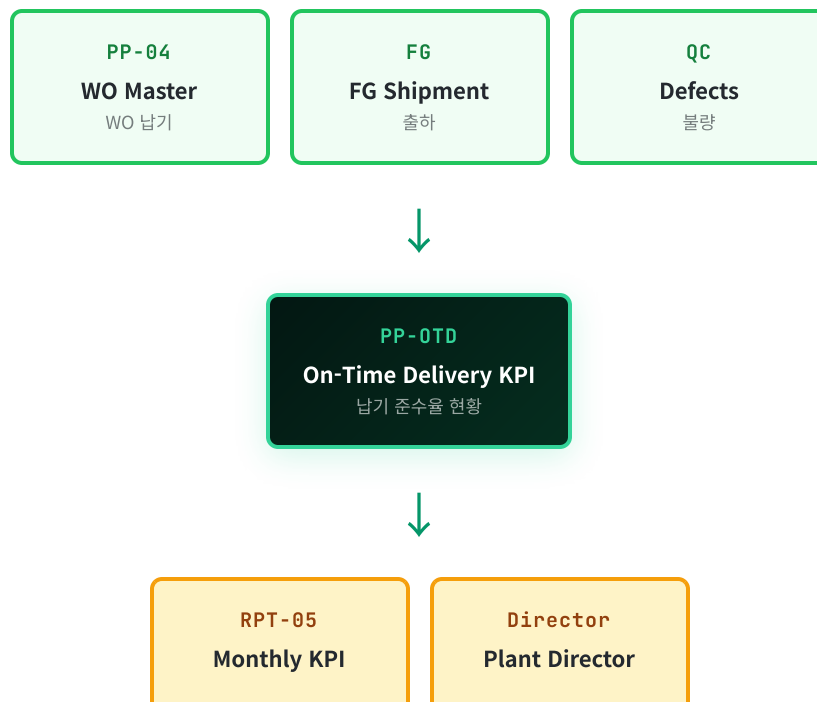
🚫 EXPORT DENIED

## 06

## Linked Screens & Data Flow

연계 화면 · 상·하류 데이터 흐름

▲ UPSTREAM · 상위 화면 / 데이터 ▲



07

# Applicable Business Rules

적용 업무 규칙

| RULE ID | RULE (EN) | 규칙 (KO) | ACTION |
|---------|-----------|---------|--------|
|---------|-----------|---------|--------|

No specific business rules apply to this screen. 이 화면에 직접 적용되는 업무 규칙은 없습니다.

A-MES · VOL.4 · PP-OTD On-Time Delivery KPI · L3 Detail Spec Automotive Parts MES · SAV & GEO Supply  
v1.0 · Bilingual EN/KO · Confidential

• VOL.4 · PP · LINE SCHEDULING · 라인 스케줄 보드

# PP-LSB LINE SCHEDULE BOARD

라인 별 분 단 위 스케줄 보드

DEVICE  Web PC

PHASE **2nd Dev**

ROLE **PLAN Manager**

FREQ **Daily · 일간**

URL **/pp/line-schedule**

## 01 Overview

화면 개요 · 목적 · 컨텍스트

A minute-level Gantt board that schedules Work Orders onto each line's **own time pattern** (MD-29). Every line row renders its active MD\_LineTimePattern as background bands — green OPERATING windows and amber PLANNED\_DOWN windows — and WO blocks (routing A blue / B orange) are dragged onto operating windows only. Available capacity per line =  $\Sigma$  OPERATING minutes; the board blocks any WO that would overflow an operating window or land on planned downtime.

작업지시(WO)를 각 라인의 **고유 시간패턴**(MD-29) 위에 배치하는 분단위 간트 보드. 각 라인 행은 활성 MD\_LineTimePattern을 배경 밴드로 렌더링한다 — 녹색 OPERATING 시간창, 호박색 PLANNED\_DOWN 시간창 — 그리고 WO 블록(라우팅 A 파랑/B 주황)은 가동 시간창에만 드래그 배치된다. 라인별 가용 캐파 =  $\Sigma$  OPERATING 분; 가동창을 초과하거나 계획 비가동에 걸치는 WO는 차단된다.

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| SCREEN ID   | PP-LSB                                                            |
| GRANULARITY | 1-minute slot · 분단위                                               |
| BASELINE    | MD-29 Line Time Pattern                                           |
| TABLES      | MD_LineTimePattern , MD_LineTimeSegment , PP_LineSchedule , PP_WO |
| RBAC        | PP.LineSchedule                                                   |
| DOWNSTREAM  | PP-ODM · PP-OEE · INJ/PNT POP                                     |

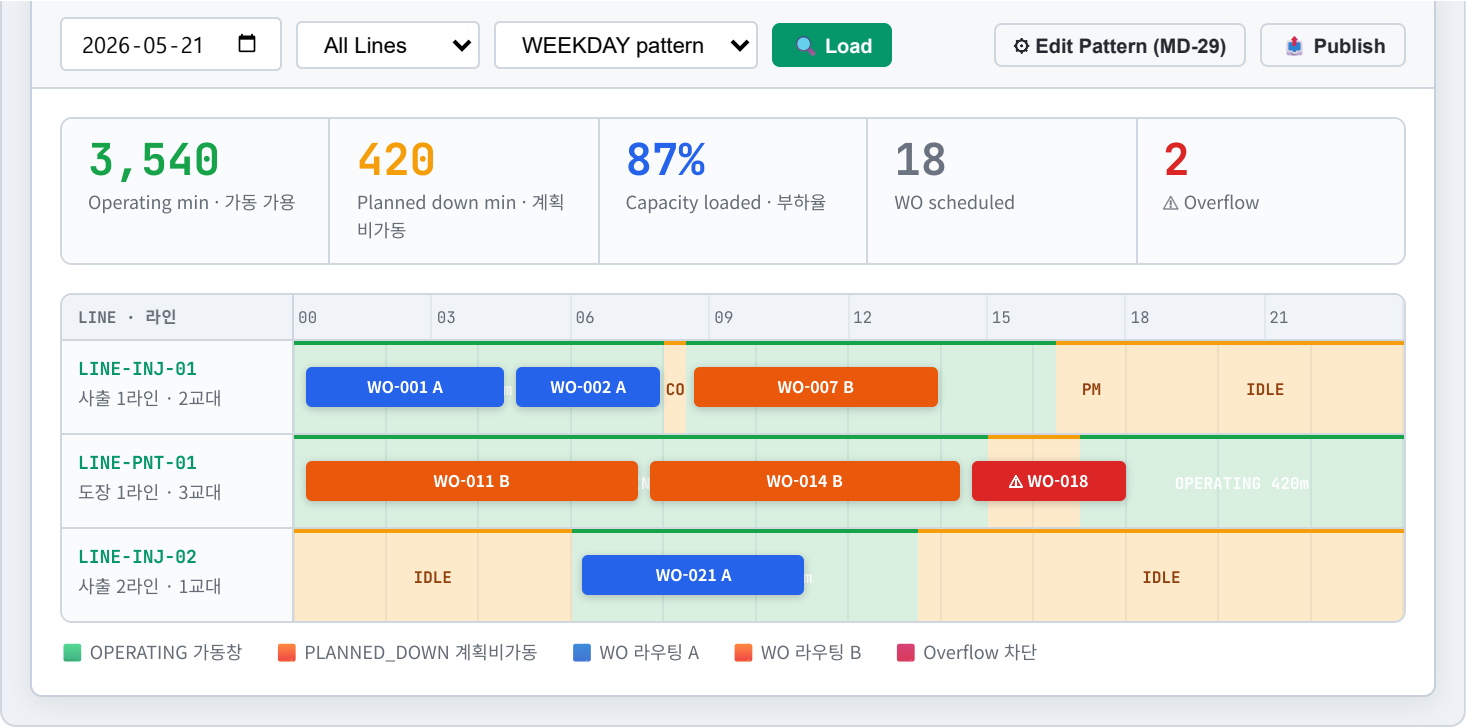
## 02 Screen Mockup

화면 목업 — 분단위 간트 보드

• PP-LSB · LINE SCHEDULE BOARD · WEB PC

 A-MES · PP · Line Schedule Board

PP > Line Schedule



03

UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트        | TYPE   | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증 · 동작                                                                                           | DEFAULT |
|--------|--------------------------|--------|------------|----------------------------------------------------------------------------------------------------------------------------|---------|
| CMP-01 | Date Picker<br>대상 일자     | DATE   | Y          | Selected date resolves each line's pattern by DayType (WEEKDAY/SAT/HOLIDAY) via MD-14 Calendar.                            | Today   |
| CMP-02 | Line Filter<br>라인 필터     | SELECT | N          | All / INJ lines / PNT lines / single line.                                                                                 | ALL     |
| CMP-03 | Pattern Band<br>시간패턴 밴드  | AUTO   | —          | Renders MD_LineTimeSegment rows: OPERATING green, PLANNED_DOWN amber. Minute-positioned via <code>StartMin/EndMin</code> . | —       |
| CMP-04 | WO Block (drag)<br>WO 블록 | DRAW   | —          | Drag a WO onto an OPERATING window. Snap to 1-min grid. Drop on PLANNED_DOWN or overflow → rejected (E-LSB-001).           | —       |
| CMP-05 | Capacity Bar<br>부하율 바    | AUTO   | —          | $Load\% = \frac{\sum WO\ minutes}{\sum OPERATING\ minutes}$ . Red >100%, amber 85-100%, green <85%.                        | —       |
| CMP-06 | Edit Pattern<br>패턴 편집    | NAV    | —          | Opens MD-29 pattern editor in modal. On save, board reloads bands. Requires <code>MD.LineTimePattern</code> role.          | —       |
| CMP-07 | Publish<br>발행            | BUTTON | N          | Commits PP_LineSchedule rows; pushes schedule to INJ/PNT POP and PP-ODM baseline. Blocked if any overflow remains.         | —       |
| CMP-08 | Auto-Schedule<br>자동 배치   | BUTTON | N          | Heuristic fill of released WOs into OPERATING windows by D-Day ascending; user reviews before Publish.                     | —       |

04

Event & Action Flow

이벤트 흐름 · 사용자 액션 → 시스템 반응

1

USER

page.load · select date

Resolves each line's DayType from MD-14 Calendar, loads the matching ACTIVE MD\_LineTimePattern and renders minute-level bands.

MD-14 캘린더로 라인별 요일유형 판별 → ACTIVE 시간패턴 로드 → 분단위 밴드 렌더링.

API

`GET /api/pp/line-schedule?date=2026-05-21`

2

USER

drag · WO block → operating window

User drags a released WO onto an OPERATING segment; system snaps to 1-minute grid and computes end minute from WO standard run time.

발행 WO를 OPERATING 세그먼트에 드래그 → 1분 그리드 스냅 → WO 표준시간으로 종료 분 계산.

CALC

`endMin = startMin + ceil(WO.qty × cycleTime / 60)`

3

SYSTEM

validate · window fit

Checks the WO block stays inside one OPERATING window and does not cross a PLANNED\_DOWN segment; overflow flips the block red.

WO 블록이 단일 OPERATING 창 내에 있고 PLANNED\_DOWN을 가로지르지 않는지 검증 — 초과 시 빨강 표시.

**RULE** BR-PP-031 · overflow → block + E-LSB-001

4

**USER** cClick · [Auto-Schedule]

Heuristically fills all unscheduled released WOs into available OPERATING minutes, ordered by D-Day ascending.

미배치 발행 WO를 D-Day 오름차순으로 가용 OPERATING 분에 자동 충전.

**API** POST /api/pp/line-schedule/auto-fill

5

**USER** cClick · [Publish]

Commits PP\_LineSchedule, pushes the day's schedule to INJ/PNT POP screens and sets the PP-ODM operating baseline.

PP\_LineSchedule 확정 → INJ/PNT POP에 일정 푸시 → PP-ODM 가동 기준선 설정.

**DOWNSTREAM** PP-ODM baseline · PP-OEE plan · POP work queue

## 05 Exception Handling

예외 처리 · 오류 코드

**E-LSB-001**

### Window Overflow

가동창 초과

|        |                                                                          |
|--------|--------------------------------------------------------------------------|
| CAUSE  | WO block exceeds the OPERATING window or crosses a PLANNED_DOWN segment. |
| DETECT | On drop / auto-fill.                                                     |
| ACTION | Block turns red, Publish disabled. Split WO or move to another window.   |

🚫 OVERFLOW · BLOCKED

**E-LSB-002**

### No Active Pattern

패턴 미정의

|        |                                                                 |
|--------|-----------------------------------------------------------------|
| CAUSE  | Line has no ACTIVE MD_LineTimePattern for the resolved DayType. |
| DETECT | On page load.                                                   |
| ACTION | Line row greyed; prompt to define pattern in MD-29.             |

🚫 PATTERN MISSING

**E-LSB-003**

### Capacity Over 100%

부하율 초과

|        |                                                                    |
|--------|--------------------------------------------------------------------|
| CAUSE  | $\Sigma$ WO minutes exceeds $\Sigma$ OPERATING minutes for a line. |
| DETECT | On every schedule change.                                          |
| ACTION | Capacity bar red; suggest moving WO to another line / day.         |

⚠ OVER-CAPACITY

#### E-LSB-004 Pattern Changed After Publish

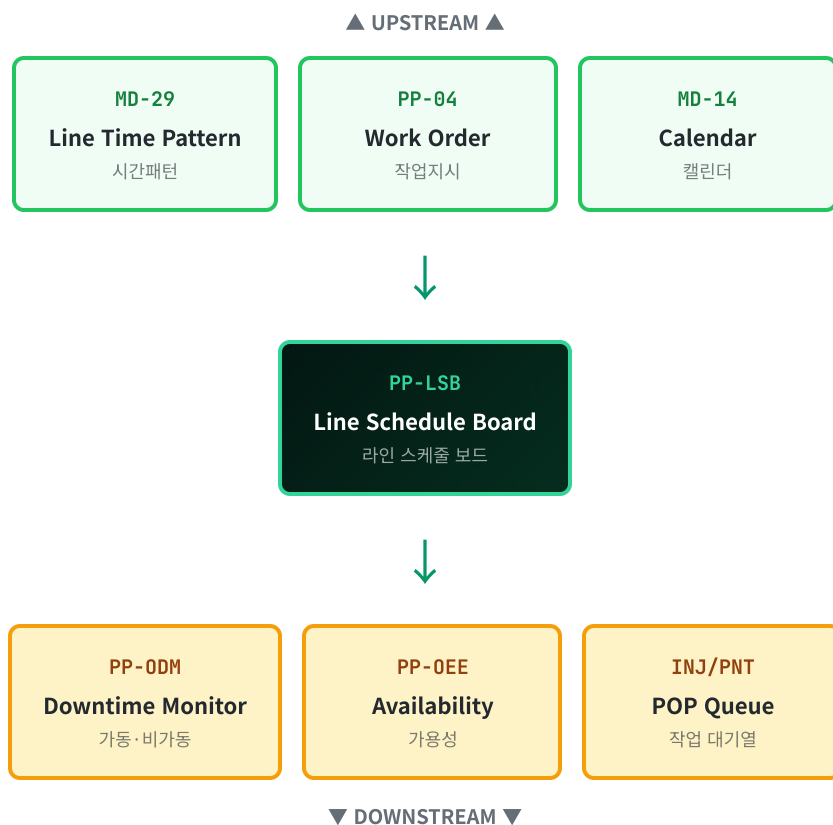
발행 후 패턴 변경

|        |                                                   |
|--------|---------------------------------------------------|
| CAUSE  | MD-29 pattern edited after the day was published. |
| DETECT | Pattern version compare.                          |
| ACTION | Banner: re-validate & re-publish required.        |

⚠ STALE SCHEDULE

## 06 Linked Screens & Data Flow

연계 화면 · 상·하류 흐름



07

Applicable Business Rules

적용 업무 규칙

| RULE ID   | RULE (EN)                                                                           | 규칙 (KO)                                            | ACTION |
|-----------|-------------------------------------------------------------------------------------|----------------------------------------------------|--------|
| BR-PP-030 | Each line schedules on its own MD-29 time pattern, not the shared factory calendar. | 각 라인은 공용 캘린더가 아닌 고유 MD-29 시간 패턴으로 스케줄링.            | AUT    |
| BR-PP-031 | A WO block must fit entirely within one OPERATING window.                           | WO 블록은 단일 OPERATING 창 안에 완전히 들어가야 함.               | BLK    |
| BR-PP-032 | Line capacity = $\Sigma$ OPERATING minutes; load over 100% blocks Publish.          | 라인 캐파 = $\Sigma$ OPERATING 분; 부하율 100% 초과 시 발행 차단. | BLK    |
| BR-PP-033 | Publish pushes the schedule to POP and sets the PP-ODM operating baseline.          | 발행 시 POP에 일정 푸시 + PP-ODM 가동 기준선 설정.                | AUT    |
| BR-PP-034 | A pattern edited after publish flags the schedule stale for re-validation.          | 발행 후 패턴 변경 시 일정을 stale 표기 — 재검증 필요.                | ALT    |





01

Overview

화면 개요 · 목적 · 컨텍스트

A real-time wallboard that compares each line's **planned time pattern** (MD-29 baseline) against its **actual operating state** streamed from equipment signals. Every minute of the day resolves to one of three states. When the plan says OPERATING but the line signal says stopped, the gap is auto-classified **UNPLANNED\_DOWN** and the operator is prompted to log a reason via PP-DTL. The classified state log feeds PP-OEE availability and RPT-06 OEE.

각 라인의 **계획 시간패턴**(MD-29 기준선)과 설비 신호로 수신되는 **실제 가동 상태**를 실시간 대조하는 현황판. 하루의 매 분(分)이 세 상태 중 하나로 판정된다. 계획상 OPERATING인데 라인 신호가 정지되면 그 구간은 **UNPLANNED\_DOWN**으로 자동 분류되고 작업자에게 PP-DTL 사유 입력을 요청한다. 분류된 상태 로그는 PP-OEE 가용성·RPT-06 OEE로 전달된다.

|            |                                                           |
|------------|-----------------------------------------------------------|
| SCREEN ID  | PP-ODM                                                    |
| REFRESH    | WebSocket · 1-minute tick                                 |
| BASELINE   | MD-29 ACTIVE pattern                                      |
| TABLES     | PP_LineStateLog , MD_LineTimePattern , PP_LineDowntimeLog |
| RBAC       | PP.DowntimeMonitor                                        |
| DOWNSTREAM | PP-DTL · PP-OEE · RPT-06                                  |

OPERATING

가동

계획 OPERATING 창에서 설비 신호가 RUN. 정상 생산 시간으로 집계되어 OEE 가용시간(Run Time)에 반영.

#### PLANNED\_DOWN

### 계획 비가동

MD-29 패턴이 PLANNED\_DOWN으로 정의한 구간 (PM·금형교체·휴게). OEE 부하시간(분모)에서 차감되는 정상 손실.

#### UNPLANNED\_DOWN

### 실제 비가동

계획상 OPERATING이나 설비 정지. 고장·자재대기·품질정지 등 — PP-DTL 사유 입력 필요, OEE 가용성 손실로 집계.

## 02 Screen Mockup

화면 목업 — 실시간 가동·비가동 현황판

● PP-ODM · OPERATING / DOWNTIME MONITOR · WEB PC 현황판

A-MES · PP · Operating / Downtime Monitor

● LIVE · 14:37

All Lines ▾

Today 2026-05-21 ▾

🔄 Auto-refresh ON

⚠ 2 Unlogged Downtime

91.4%

Operating rate · 가동률

320

Planned down min · 계획  
비가동

74

Unplanned down min ·  
실제비가동

2

⚠ Reason unlogged

5 / 6

Lines running

| LINE · STATE                | 06          | 09 | 12 | 15 | 18 | 21 |
|-----------------------------|-------------|----|----|----|----|----|
| LINE-INJ-01<br>● OPERATING  | PLAN · 계획   |    |    |    |    |    |
|                             | ACTUAL · 실제 |    |    |    |    |    |
|                             |             |    |    |    |    |    |
| LINE-PNT-01<br>● UNPLANNED  | PLAN · 계획   |    |    |    |    |    |
|                             | ACTUAL · 실제 |    |    |    |    |    |
|                             |             |    |    |    |    |    |
| LINE-INJ-02<br>● PLANNED DN | PLAN · 계획   |    |    |    |    |    |
|                             | ACTUAL · 실제 |    |    |    |    |    |
|                             |             |    |    |    |    |    |

| Time        | Line        | State     | Duration | Reason · 사유      | Action |
|-------------|-------------|-----------|----------|------------------|--------|
| 13:58-14:09 | LINE-PNT-01 | UNPLANNED | 11 min   | △ 미입력            | 사유 입력  |
| 14:24-14:37 | LINE-PNT-01 | UNPLANNED | 13 min ⌚ | △ 진행중            | 사유 입력  |
| 12:00-12:48 | LINE-INJ-02 | PLANNED   | 48 min   | RSN-DT-PM · 예방보전 | 정상     |

## 03 UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트                        | TYPE   | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증 · 동작                                                                                                        | DEFAULT |
|--------|------------------------------------------|--------|------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------|
| CMP-01 | <b>Plan vs Actual Lane</b><br>계획 · 실제 레인 | AUTO   | —          | Two stacked minute bands per line — PLAN from MD-29, ACTUAL from PP_LineStateLog. NOW vertical line marks current minute.               | —       |
| CMP-02 | <b>State Classifier</b><br>상태 분류기        | AUTO   | —          | Per minute: signal RUN + plan OPERATING → OPERATING · plan PLANNED_DOWN → PLANNED_DOWN · signal STOP + plan OPERATING → UNPLANNED_DOWN. | —       |
| CMP-03 | <b>Operating Rate KPI</b><br>가동률         | AUTO   | —          | $\text{Rate} = \frac{\text{OPERATING min}}{\text{OPERATING} + \text{UNPLANNED\_DOWN min}}$ elapsed today.                               | —       |
| CMP-04 | <b>Unlogged Downtime Alert</b><br>미입력 경보 | AUTO   | —          | Counts UNPLANNED_DOWN events with no PP_LineDowntimeLog reason. Red badge + row highlight.                                              | 0       |
| CMP-05 | <b>Log Reason Button</b><br>사유 입력        | NAV    | —          | Opens PP-DTL pre-filled with line, start/end minute and pattern reference.                                                              | —       |
| CMP-06 | <b>Line Filter</b><br>라인 필터              | SELECT | N          | All / by process / single line.                                                                                                         | ALL     |
| CMP-07 | <b>Auto-refresh</b><br>자동 갱신             | TOGGLE | N          | WebSocket 1-minute tick. When off, manual refresh only.                                                                                 | ON      |

## 04 Event & Action Flow

이벤트 흐름 · 상태 분류 → 사유 입력

1

**TICK** minute.tick · equipment signal

Every minute, equipment RUN/STOP signal (via SYS-04 interface) is paired with the line's MD-29 planned state for that minute.

매 분 설비 RUN/STOP 신호(SYS-04 인터페이스)를 해당 분의 MD-29 계획 상태와 짝지음.

```
IN signal{lined, minute, runFlag} + MD_LineTimeSegment.state
```

2

#### SYSTEM classify · 3-state

Resolves the minute to OPERATING / PLANNED\_DOWN / UNPLANNED\_DOWN and appends a PP\_LineStateLog row.

해당 분을 OPERATING / PLANNED\_DOWN / UNPLANNED\_DOWN으로 판정 → PP\_LineStateLog 행 추가.

```
RULE BR-PP-035 · STOP within OPERATING window → UNPLANNED_DOWN
```

3

#### SYSTEM detect · downtime event

Consecutive UNPLANNED\_DOWN minutes are merged into one open downtime event; the board badges it as reason-unlogged.

연속 UNPLANNED\_DOWN 분(分)을 하나의 진행중 비가동 이벤트로 병합 — 사유 미입력 배지 표시.

```
API POST /api/pp/downtime-event · status=OPEN
```

4

#### SYSTEM alert · downtime threshold

An unplanned downtime exceeding 10 minutes pushes an alert to PLAN / MNT supervisors.

실제 비가동 10분 초과 시 PLAN/MNT 감독자에게 경보 푸시.

```
ALERT BR-PP-037 · in-app + SYS-06 notification
```

5

#### USER click · [사유 입력] → PP-DTL

Operator logs a downtime reason; the event closes and the classified state log flows to PP-OEE availability.

작업자가 비가동 사유 입력 → 이벤트 종료 → 분류 상태 로그가 PP-OEE 가용성으로 전달.

```
DOWNSTREAM PP-DTL close · PP-OEE recalc · RPT-06 OEE
```

## 05 Exception Handling

예외 처리 · 오류 코드

### E-ODM-001 Signal Loss

신호 두절

|        |                                                               |
|--------|---------------------------------------------------------------|
| CAUSE  | No equipment RUN/STOP signal received for > 3 minutes.        |
| DETECT | Heartbeat timeout.                                            |
| ACTION | Lane greyed as UNKNOWN; minutes held pending, not classified. |

— SIGNAL LOST

### E-ODM-002 Reason Unlogged

사유 미입력

CAUSE UNPLANNED\_DOWN event closed without a reason for > 30 min.

DETECT Scheduled sweep.

ACTION Escalation alert to supervisor; event stays OPEN.

⚠ REASON OVERDUE

E-ODM-003

### Run During Planned Down

계획비가동 중 가동

CAUSE Equipment RUN while plan says PLANNED\_DOWN.

DETECT Classifier mismatch.

ACTION Counted as OPERATING (extra run); flagged for pattern review.

⚠ UNPLANNED RUN

E-ODM-004

### No Pattern Baseline

기준 패턴 없음

CAUSE Line has no ACTIVE MD-29 pattern for today.

DETECT On monitor load.

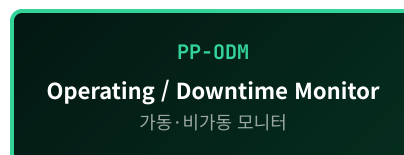
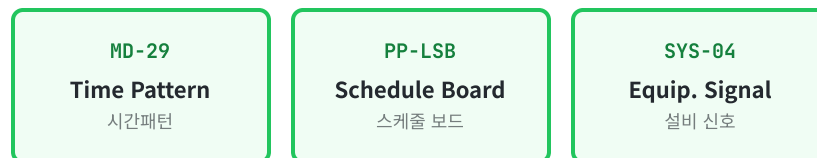
ACTION Cannot classify; lane shows 'define pattern' prompt.

⊖ BASELINE MISSING

## 06 Linked Screens & Data Flow

연계 화면 · 상·하류 흐름

▲ UPSTREAM ▲





▼ DOWNSTREAM ▼

## 07

### Applicable Business Rules

적용 업무 규칙

| RULE ID   | RULE (EN)                                                                            | 규칙 (KO)                                        | ACTION |
|-----------|--------------------------------------------------------------------------------------|------------------------------------------------|--------|
| BR-PP-035 | Equipment STOP inside a planned OPERATING window is auto-classified UNPLANNED_DOWN.  | 계획 OPERATING 창 내 설비 정지는 UNPLANNED_DOWN 자동 분류.  | AUT    |
| BR-PP-036 | Every UNPLANNED_DOWN event requires a reason code logged via PP-DTL.                 | 모든 실제 비가동 이벤트는 PP-DTL 사유코드 입력 필수.              | BLK    |
| BR-PP-037 | Unplanned downtime over 10 minutes escalates an alert to PLAN / MNT.                 | 실제 비가동 10분 초과 시 PLAN/MNT 경보에 스칼레이션.            | ALT    |
| BR-PP-038 | Operating rate = OPERATING / (OPERATING + UNPLANNED_DOWN) excludes planned downtime. | 가동률 = 가동 / (가동 + 실제비가동) — 계획비가동 제외.            | AUT    |
| BR-PP-039 | The classified PP_LineStateLog is the single source for PP-OEE and RPT-06.           | 분류된 PP_LineStateLog가 PP-OEE·RPT-06의 단일 기준 데이터. | AUT    |

• VOL.4 · PP · LINE SCHEDULING · 라인 가용성 · OEE

# PP-OEE LINE AVAILABILITY & OEE

라 인 가 용 성 · O E E 롤 업

DEVICE  Web PC

PHASE 2nd Dev

ROLE PLAN · MGT

FREQ Daily / Period

URL /pp/line-oee

## 01 Overview

화면 개요 · 목적 · 컨텍스트

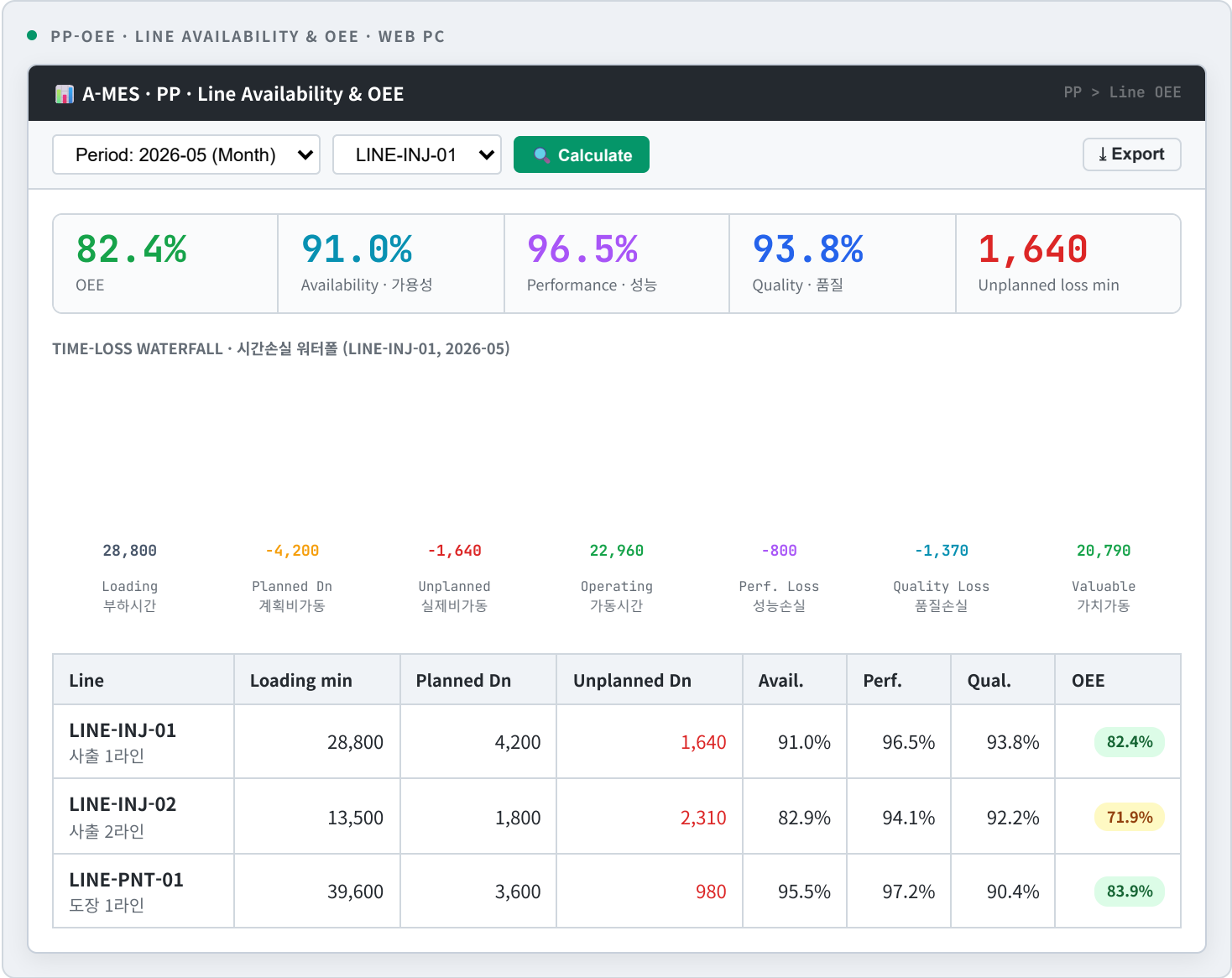
Rolls up the classified PP\_LineStateLog into a time-loss model and computes **OEE** per line and period. Loading Time ( $\Sigma$  planned OPERATING from MD-29) minus PLANNED\_DOWN and UNPLANNED\_DOWN yields the Operating Time and **Availability**; combined with Performance (cycle-time efficiency) and Quality (QC yield) it produces OEE. The screen separates planned vs unplanned loss so each line's own time pattern is fairly credited.

분류된 PP\_LineStateLog를 시간손실 모델로 롤업하여 라인·기간별 **OEE**를 산출한다. 부하시간(MD-29 계획 OPERATING 합계)에서 PLANNED\_DOWN·UNPLANNED\_DOWN을 차감해 가동시간과 **가용성**을 구하고, 성능(사이클타임 효율)·품질(QC 수율)과 결합해 OEE를 산출한다. 계획/비계획 손실을 분리 표기하여 각 라인 고유 시간패턴이 공정하게 반영된다.

|            |                                                   |
|------------|---------------------------------------------------|
| SCREEN ID  | PP-OEE                                            |
| PERIOD     | Day / Week / Month / custom                       |
| SOURCE     | PP_LineStateLog (PP-ODM)                          |
| TABLES     | PP_LineStateLog , PP_LineOEE , MD_LineTimePattern |
| RBAC       | PP.LineOEE                                        |
| DOWNSTREAM | RPT-06 OEE · RPT-07 KPI                           |

$\text{Loading Time} = \Sigma \text{ MD-29 OPERATING minutes}$  |  $\text{Operating Time} = \text{Loading} - \text{Planned Down} - \text{Unplanned Down}$   
 $\text{Availability} = \text{Operating Time} \div \text{Loading Time}$       $\text{Performance} = (\text{Ideal cycle} \times \text{Qty}) \div \text{Operating Time}$   
 $\text{Quality} = \text{Good Qty} \div \text{Total Qty} \rightarrow \text{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality}$

## 02 Screen Mockup

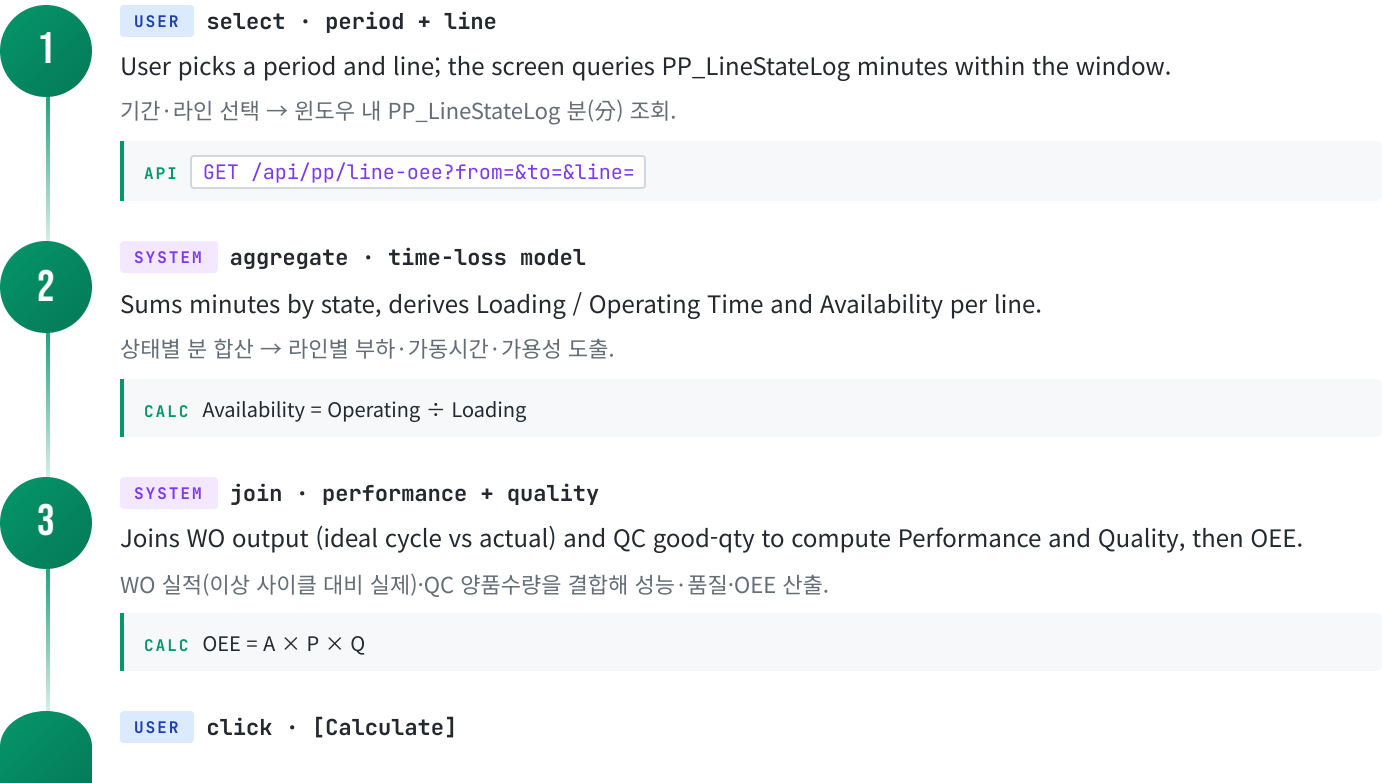




| ID     | COMPONENT<br>컴포넌트             | TYPE   | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증 · 동작                                                                                                                                                           | DEFAULT    |
|--------|-------------------------------|--------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| CMP-01 | Period Selector<br>기간 선택      | SELECT | Y          | Day / Week / Month / custom range. Drives the rollup window.                                                                                                                               | This Month |
| CMP-02 | Line Selector<br>라인 선택        | SELECT | N          | Single line drives the waterfall; All lines drives the rollup table.                                                                                                                       | All        |
| CMP-03 | OEE KPI Tiles<br>OEE KPI      | AUTO   | -          | OEE / Availability / Performance / Quality. Green $\geq$ 85%, amber 70-85%, red $<$ 70%.                                                                                                   | -          |
| CMP-04 | Time-loss Waterfall<br>손실 워터폴 | AUTO   | -          | Loading $\rightarrow$ Planned Dn $\rightarrow$ Unplanned Dn $\rightarrow$ Operating $\rightarrow$ Perf. loss $\rightarrow$ Quality loss $\rightarrow$ Valuable. Bars from PP_LineStateLog. | -          |
| CMP-05 | Line Rollup Table<br>라인 롤업 표  | AUTO   | -          | One row per line with the full loss breakdown and OEE; sortable.                                                                                                                           | OEE desc   |
| CMP-06 | Calculate<br>산출               | BUTTON | N          | Recomputes the rollup; persists a PP_LineOEE snapshot for the period.                                                                                                                      | -          |
| CMP-07 | Export<br>내보내기                | BUTTON | N          | Exports to Excel / PDF; feeds RPT-06 OEE and RPT-07 monthly KPI.                                                                                                                           | -          |

## 04 Event & Action Flow

이벤트 흐름 · 롤업 산출





Persists a PP\_LineOEE snapshot row per line for the period so the figure is auditable and stable.

기간별 라인 PP\_LineOEE 스냅샷 저장 — 수치 감사·고정.

API POST /api/pp/line-oe/snapshot

SCHEDULED nightly · auto-rollup

A nightly job snapshots the prior day's OEE and pushes it to RPT-06 and RPT-07.

야간 배치가 전일 OEE를 스냅샷하여 RPT-06·RPT-07로 전달.

DOWNSTREAM RPT-06 Equipment OEE · RPT-07 Monthly KPI

## 05

### Exception Handling

예외 처리 · 오류 코드

E-OEE-001

#### Unlogged Downtime in Period

미입력 비가동 포함

CAUSE Period contains UNPLANNED\_DOWN events with no reason logged.

DETECT On calculate.

ACTION OEE shown as provisional; banner links to PP-DTL backlog.

⚠ PROVISIONAL OEE

E-OEE-002

#### No State Log

상태 로그 없음

CAUSE PP\_LineStateLog empty for the selected window (PP-ODM not run).

DETECT On query.

ACTION Cannot compute; prompt to verify PP-ODM coverage.

⊖ NO DATA

E-OEE-003

#### Signal-Loss Minutes

신호두절 구간

CAUSE UNKNOWN-state minutes (E-ODM-001) inside the period.

DETECT Aggregate scan.

ACTION Excluded from Loading; coverage % shown beside OEE.

⚠ PARTIAL COVERAGE

E-OEE-004

### OEE Below Target

OEE 목표 미달

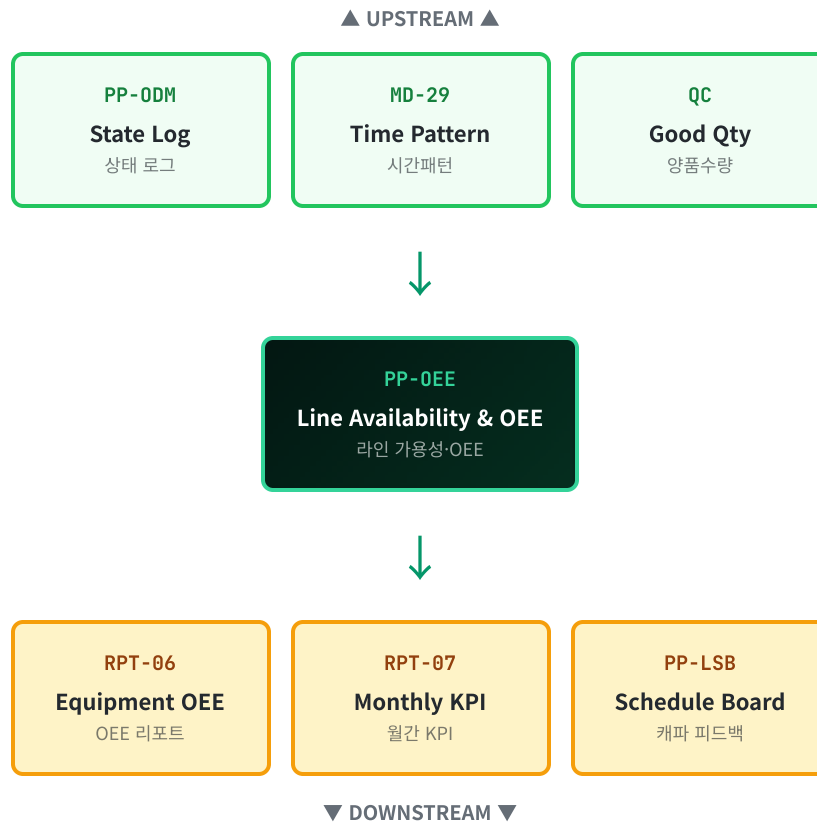
|        |                                                 |
|--------|-------------------------------------------------|
| CAUSE  | Line OEE under the 75% target.                  |
| DETECT | On calculate.                                   |
| ACTION | Row red; alert to PLAN manager for loss review. |

⚠ OEE < TARGET

## 06

### Linked Screens & Data Flow

연계 화면 · 상·하류 흐름



## 07

### Applicable Business Rules

적용 업무 규칙

| RULE ID   | RULE (EN)                                                                          | 규칙 (KO)                                        | ACTION |
|-----------|------------------------------------------------------------------------------------|------------------------------------------------|--------|
| BR-PP-040 | Loading Time = $\Sigma$ MD-29 OPERATING minutes for the line and period.           | 부하시간 = 라인·기간의 MD-29 OPERATING 분 합계.            | AUT    |
| BR-PP-041 | Availability counts only unplanned downtime as loss; planned downtime is excluded. | 가용성은 실제 비가동만 손실로 집계 — 계획 비가동 제외.               | AUT    |
| BR-PP-042 | OEE = Availability $\times$ Performance $\times$ Quality, snapshotted per period.  | OEE = 가용성 $\times$ 성능 $\times$ 품질, 기간별 스냅샷 저장. | AUT    |
| BR-PP-043 | A period with unlogged downtime yields a provisional (non-final) OEE.              | 미입력 비가동 포함 기간은 잠정(비확정) OEE로 표기.                | ALT    |
| BR-PP-044 | Line OEE below the 75% target alerts the PLAN manager for loss review.             | 라인 OEE 75% 목표 미달 시 PLAN 관리자 손실 검토 경보.          | ALT    |

• VOL.4 · PP · LINE SCHEDULING · 비가동 로그 입력

# PP-DTL DOWNTIME LOG ENTRY

비 가 동 로 그 입 력 · 사 유 기 록

DEVICE  POP 터치패널 / PDA

PHASE 2nd Dev

ROLE Line Operator

FREQ On event · 이벤트

URL /pop/downtime-log

## 01 Overview

화면 개요 · 목적 · 컨텍스트

A POP touch-panel screen where the line operator records the **reason for an unplanned downtime event** detected by PP-ODM. The event arrives pre-filled with line, start/end minute and live duration; the operator taps a downtime **reason code** (MD-25, filtered to ReasonType=DOWNTIME) and an optional 4M1E **cause** (MD-22), enters a comment if the reason requires it, and confirms. The closed event with its reason flows to PP-ODM and PP-OEE so availability loss is correctly attributed.

PP-ODM이 감지한 **실제 비가동 이벤트의 사유**를 라인 작업자가 기록하는 POP 터치패널 화면. 이벤트는 라인·시작/종료 분·실시간 경과시간이 미리 채워져 도착하며, 작업자는 비가동 **사유코드**(MD-25, ReasonType=DOWNTIME 필터)와 선택적 4M1E **원인**(MD-22)을 터치 선택하고, 사유가 요구하면 코멘트를 입력한 뒤 확정한다. 사유가 기록된 종료 이벤트는 PP-ODM·PP-OEE로 전달되어 가용성 손실이 정확히 귀속된다.

|            |                                      |
|------------|--------------------------------------|
| SCREEN ID  | PP-DTL                               |
| DEVICE     | POP 15" touch · PDA                  |
| REASON SRC | MD-25 (DOWNTIME) · MD-22             |
| TABLES     | PP_LineDowntimeLog , PP_LineStateLog |
| RBAC       | PP.DowntimeLog                       |
| DOWNSTREAM | PP-ODM · PP-OEE · MNT-06             |

## 02 Screen Mockup

화면 목업 — POP 터치패널

⚠ UNPLANNED DOWN

## LINE-PNT-01 · 도장 1라인

14:24 정지 → 진행중 · 계획상 OPERATING 구간

00:13:24

## ① 비가동 사유 · REASON (MD-25)



설비 고장

RSN-DT-BRK



자재 대기

RSN-DT-MAT



품질 정지

RSN-DT-QC



금형 트러블

RSN-DT-MLD

## ② 원인 (4M1E) · CAUSE (MD-22)

MATERIAL

자재

METHOD

방법

MAN

사람

💬 코멘트 (필수) — 분체 로트 교체 지연, 입고 대기 ...

취소

✓ 사유 확정 · 이벤트 종료

터치 1·2단계 선택 → 코멘트 → 확정. 진행중 이벤트는 종료 시 EndMin 자동 확정.

## 03

## UI Components Specification

UI 컴포넌트 명세 · 검증 규칙

| ID     | COMPONENT<br>컴포넌트              | TYPE   | REQ.<br>필수 | VALIDATION / BEHAVIOR<br>검증 · 동작                                                                                 | DEFAULT  |
|--------|--------------------------------|--------|------------|------------------------------------------------------------------------------------------------------------------|----------|
| CMP-01 | Event Card<br>이벤트 카드           | AUTO   | —          | Shows line, start minute, live duration. Open events tick every second until reason confirmed.                   | —        |
| CMP-02 | Reason Tiles<br>사유 타일          | TOUCH  | Y          | Big touch tiles loaded from MD-25 where ReasonType=DOWNTIME and AppliesToModule covers the line's process.       | (none)   |
| CMP-03 | Cause Tiles<br>(4M1E)<br>원인 타일 | TOUCH  | N          | Optional MD-22 cause; tiles filter by the selected reason's process scope.                                       | (none)   |
| CMP-04 | Comment<br>Field<br>코멘트        | TEXT   | —          | Becomes required when the chosen reason has <b>RequiresComment=TRUE</b> ; on-screen keyboard.                    | (empty)  |
| CMP-05 | Confirm<br>Button<br>확정        | BUTTON | Y          | Disabled until a reason is picked and any required comment entered. Writes PP_LineDowntimeLog, closes the event. | Disabled |
| CMP-06 | End-time<br>Capture<br>종료시각    | AUTO   | —          | For an in-progress event, EndMin is set to the equipment RESUME minute or the confirm minute, whichever first.   | —        |
| CMP-07 | Pending Queue<br>미입력 대기        | LIST   | —          | Lists all unlogged downtime events for the line so the operator can clear the backlog.                           | —        |

04

Event & Action Flow

이벤트 흐름 · 사유 입력 → 종료

1

SYSTEM

push · downtime event open

PP-ODM detects an unplanned downtime and pushes an open event to the line's POP panel.

PP-ODM이 실제 비가동을 감지하여 라인 POP 패널에 진행중 이벤트를 푸시.

PUSH

WS event: `downtime.open {lineId, startMin}`

2

USER

tap · reason + cause

Operator taps a DOWNTIME reason tile (MD-25) and, if needed, a 4M1E cause tile (MD-22).

작업자가 DOWNTIME 사유 타일(MD-25)과 필요 시 4M1E 원인 타일(MD-22)을 터치.

STATE

reason set · cause optional

3

**SYSTEM validate · comment requirement**

If the chosen reason has RequiresComment=TRUE, the comment field is enforced before Confirm enables.  
선택 사유가 코멘트 필수면 Confirm 활성화 전 코멘트 입력 강제.

**RULE** BR-PP-046 · MD\_ReasonCode.RequiresComment

4

**USER tap · [사유 확정]**

Writes a PP\_LineDowntimeLog row (line, start/end minute, reason, cause, comment, operator) and closes the event.

PP\_LineDowntimeLog 행 기록(라인 · 시작/종료분 · 사유 · 원인 · 코멘트 · 작업자) → 이벤트 종료.

**API** POST /api/pp/downtime-log

5

**SYSTEM complete · attribute loss**

The reasoned downtime updates PP-ODM (event cleared), feeds PP-OEE availability loss and mirrors to MNT-06 if reason is a breakdown.

사유 기록된 비가동이 PP-ODM 갱신(이벤트 해소)·PP-OEE 가용성 손실 반영 · 고장 사유면 MNT-06 동기화.

**DOWNSTREAM** PP-ODM clear · PP-OEE recalc · MNT-06 downtime

## 05 Exception Handling

예외 처리 · 오류 코드

**E-DTL-001****Confirm Without Reason**

사유 미선택 확정

|        |                                                |
|--------|------------------------------------------------|
| CAUSE  | Operator taps Confirm with no reason selected. |
| DETECT | Confirm guard.                                 |
| ACTION | Confirm stays disabled; reason tiles pulse.    |

REASON REQUIRED

**E-DTL-002****Comment Missing**

코멘트 누락

|        |                                               |
|--------|-----------------------------------------------|
| CAUSE  | Reason requires a comment but field is empty. |
| DETECT | On confirm.                                   |
| ACTION | Comment field highlighted; keyboard opens.    |

COMMENT REQUIRED



E-DTL-003

### Event Auto-Resumed

자동 재가동

|        |                                                                      |
|--------|----------------------------------------------------------------------|
| CAUSE  | Equipment resumes while operator is still picking a reason.          |
| DETECT | Signal RESUME.                                                       |
| ACTION | EndMin frozen at resume; reason entry continues for the closed span. |

⚠ AUTO-CLOSED · LOG STILL DUE

E-DTL-004

### Offline POP

POP 오프라인

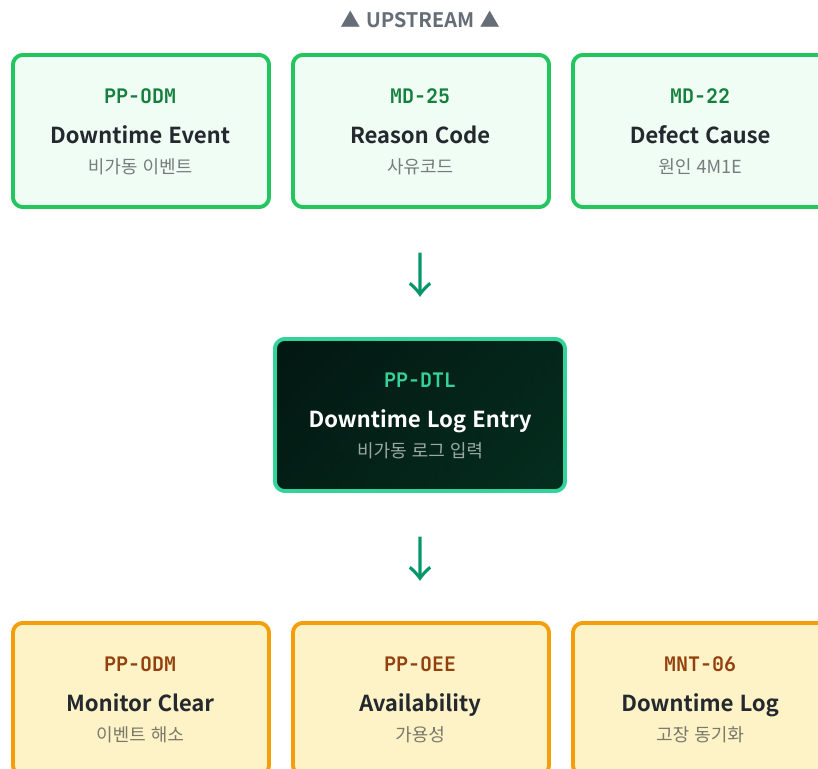
|        |                                               |
|--------|-----------------------------------------------|
| CAUSE  | Network drop while confirming.                |
| DETECT | API timeout.                                  |
| ACTION | Log queued locally, auto-synced on reconnect. |

🛑 OFFLINE · QUEUED

## 06

## Linked Screens & Data Flow

연계 화면 · 상·하류 흐름



## 07

## Applicable Business Rules

적용 업무 규칙

| RULE ID   | RULE (EN)                                                                     | 규칙 (KO)                                | ACTION |
|-----------|-------------------------------------------------------------------------------|----------------------------------------|--------|
| BR-PP-045 | Every unplanned downtime event must be closed with a DOWNTIME reason code.    | 모든 실제 비가동 이벤트는 DOWNTIME 사유 코드로 종료해야 함. | BLK    |
| BR-PP-046 | A reason with RequiresComment=TRUE blocks confirm until a comment is entered. | 코멘트 필수 사유는 코멘트 입력 전 확정 차단.             | BLK    |
| BR-PP-047 | An in-progress event's EndMin is fixed at the equipment resume minute.        | 진행중 이벤트의 종료 분은 설비 재가동 시각으로 확정.         | AUT    |
| BR-PP-048 | A breakdown-type reason mirrors the event to the MNT-06 downtime log.         | 고장 유형 사유는 이벤트를 MNT-06 비가동 로그에 동기화.     | AUT    |
| BR-PP-049 | Offline POP entries are queued locally and auto-synced on reconnect.          | 오프라인 POP 입력은 로컬 큐 저장 후 재접속 시 자동 동기화.   | AUT    |